United States Patent	12398779
Kind Code	B2
Date of Patent	August 26, 2025
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Mechanical-energy storage unit system

Abstract

A system may include a massive flywheel including a rotatable mass component and one or more axles coupled with the rotatable mass component, the one or more axles extending from a top of the rotatable mass component and from a bottom of the rotatable mass component. A system may include a bottom bearing assembly coupled with the one or more axles at the bottom of the rotatable mass component. A system may include a top bearing assembly coupled with the one or more axles at the top of the rotatable mass component. A system may include a support structure coupled with the top bearing assembly and the bottom bearing assembly. A system may include a motor coupled with the one or more axles at the top bearing assembly.

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Appl. No.: **18/666522**

Filed: **May 16, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240388164 A1	Nov. 21, 2024

Related U.S. Application Data

us-provisional-application US 63502648 20230516

Publication Classification

Int. Cl.: **F16F15/315** (20060101); **F03G3/08** (20060101); **F16C17/02** (20060101); **H02K7/09** (20060101); H02K7/02 (20060101)

U.S. Cl.:

CPC **F16F15/3156** (20130101); **F03G3/08** (20130101); **F16C17/02** (20130101); **F16F15/3153** (20130101); **H02K7/09** (20130101); F05B2230/608 (20130101); F05B2240/40 (20130101); F05B2260/421 (20130101); F16C2361/55 (20130101); H02K7/025 (20130101)

Field of Classification Search

CPC: H02K (7/025); H02K (7/02); H02J (15/007); F16F (15/3156); F16F (15/315)

USPC: 310/74

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application No. 63/502,648 filed on May 16, 2023. The present application is related to U.S. application Ser. No. 18/666,542, titled “Flywheel Vacuum Enclosure and Adjustment System” filed on May 16, 2024; U.S. application Ser. No. 18/666,557, titled “Stacking Flywheel and Linkage” filed on May 16, 2024; U.S. application Ser. No. 18/666,573, titled “Flywheel Magnetic Lift and Bearing System” filed on May 16, 2024; and U.S. application Ser. No. 18/666,593, titled “Mechanical-Energy Storage Unit and Assembly Fixture” filed on May 16, 2024; as the present application by common inventors. All of these applications are incorporated herein by reference, including their specifications and drawings, which disclosure is not admitted to be prior art with respect to the present invention by its mention in the cross-reference section.

BACKGROUND
(1) The present disclosure relates to mechanical energy storage units. Implementations relate to flywheel-based mechanical energy storage units.
(2) Currently, residential electricity customers, as well as electrical utilities, use various sources of electrical energy storage to offset varying electrical power production and use, such as the duck curve associated with solar or other renewable energy production. The variation in power production and usage has been further exacerbated with the increasing popularity of renewable power sources. These issues cause significant cost and other issues to utilities, power outages, and other issues.
(3) Commonly, excess or backup power is stored in chemical storage, such as large chemical batteries. Unfortunately, chemical batteries suffer from many issues that make them undesirable at both a residential level and at a utility level. For example, chemical batteries may be very expensive, complex, and require numerous safeguards against fires. Chemical batteries are also ecologically unfriendly, as their production uses toxic chemicals, creates significant greenhouse gases, and results in significant material waste. Furthermore, chemical batteries have short lifespans because the batteries have a limited number of years and recharge cycles before they must be disposed of.
(4) Previous solutions for mechanical energy storage have been overly complex, too large to be implemented at a residential level, not scalable for an electrical utility, or have faced other issues.
SUMMARY
(5) In some aspects, the techniques described herein relate to a system including: a massive flywheel including a rotatable mass component and one or more axles coupled with the rotatable mass component, the one or more axles extending from a top of the rotatable mass component and from a bottom of the rotatable mass component; a bottom bearing assembly coupled with the one or more axles at the bottom of the rotatable mass component; a top bearing assembly coupled with the one or more axles at the top of the rotatable mass component; a support structure coupled with the top bearing assembly and the bottom bearing assembly; and a motor coupled with the one or more axles at the top bearing assembly.
(6) In some aspects, the techniques described herein relate to a system, wherein: the motor is mounted to the support structure via one or more braces, the one or more braces providing a space between the support structure and the motor, the space being large enough to fit a top bearing between the motor and the support structure.
(7) In some aspects, the techniques described herein relate to a system, wherein the support structure includes an enclosure having a cylindrical shape, the motor being mounted to the enclosure in line with an axis of rotation of the one or more axles.
(8) In some aspects, the techniques described herein relate to a system, wherein: the top bearing assembly coupled with the one or more axles, the top bearing assembly including a top bearing coupling with the one or more axles and a magnet that applies a pulling force to the rotatable mass component.
(9) In some aspects, the techniques described herein relate to a system, wherein the bottom bearing assembly includes a positioning mechanism configured to raise and lower a bottom bearing, the bottom bearing coupling with the one or more axles.
(10) In some aspects, the techniques described herein relate to a system, wherein the support structure includes: an enclosure tub having a bottom and one or more sides coupled with the bottom, the bottom including a perforation, the bottom bearing assembly being located inside the perforation in the bottom.
(11) In some aspects, the techniques described herein relate to a system, further including: an enclosure lid coupled with the one or more sides to form a top of the enclosure tub, the enclosure lid providing vertical support to the top bearing assembly.
(12) In some aspects, the techniques described herein relate to a system, wherein the top bearing assembly supports more than half of a weight of the rotatable mass component.
(13) In some aspects, the techniques described herein relate to a system, wherein the rotatable mass component includes a plurality of stacked plates.
(14) In some aspects, the techniques described herein relate to a system, wherein the plurality of stacked plates includes two clamping plates and two or more stacking plates located between the two clamping plates, the two clamping plates being pulled together by one or more fasteners to place a compressive force on the two or more stacking plates.
(15) In some aspects, the techniques described herein relate to a system, wherein the one or more axles include a top axle and a bottom axle, the top axle being separated from the bottom axle by the two or more stacking plates.
(16) In some aspects, the techniques described herein relate to a system, wherein: the support structure includes a base and a lid, the bottom bearing assembly being coupled with the base and the top bearing assembly being coupled with the lid; the motor is coupled with the lid via one or more braces; and the lid includes an accessory mounting plate holding one or more of a vacuum assembly, a supercapacitor, and a central processing unit.
(17) In some aspects, the techniques described herein relate to a system, wherein the lid includes a plurality of ribs separating a body of the lid

from the accessory mounting plate, the plurality of ribs merging at a ring at a center of the lid, at least a portion of the top bearing assembly being located within a radius of the ring.
(18) In some aspects, the techniques described herein relate to a system, further including: a plurality of magnets mounted to a bottom surface of the lid, the plurality of magnets lifting the rotatable mass component to remove a vertical force from the bottom bearing assembly.
(19) In some aspects, the techniques described herein relate to a system, wherein the support structure includes a plurality of feet around a peripheral edge of the support structure.
(20) In some aspects, the techniques described herein relate to a system, wherein: the support structure includes a cylindrical enclosure, the rotatable mass component being located inside the cylindrical enclosure; the motor and a vacuum assembly are mounted on top of the support structure; and a plurality of feet are located around a peripheral edge of the cylindrical enclosure.
(21) In some aspects, the techniques described herein relate to a mechanical-energy storage unit including: a massive flywheel including two clamping plates and two or more stacking plates located between the two clamping plates, the two clamping plates being pulled together by one or more fasteners to place a compressive force on the two or more stacking plates; a top axle coupled with a top clamping plate of the two clamping plates, a bottom axle coupled with a bottom clamping plate of the two clamping plates; a top bearing coupled with the top axle; a bottom bearing coupled with the bottom axle; and a support structure coupled with the top bearing and the bottom bearing.
(22) In some aspects, the techniques described herein relate to a mechanical-energy storage unit, wherein: the top axle is separated from the bottom axle by the two or more stacking plates; the top axle is held at an axis of rotation by the top bearing; and the bottom axle is held at the axis of rotation by the bottom bearing.
(23) In some aspects, the techniques described herein relate to a mechanical-energy storage unit, further including: a plurality of magnets mounted the support structure, the plurality of magnets lifting the massive flywheel toward the top bearing.
(24) In some aspects, the techniques described herein relate to a flywheel assembly including: an enclosure having a top, a bottom, and an interior cavity; a massive flywheel located in the interior cavity of the enclosure; a bottom axle coupled with a bottom side of the massive flywheel; a top axle coupled with a top side of the massive flywheel, the top axle being a separate component than the bottom axle; a bottom bearing coupling the bottom axle with the bottom of the enclosure; and a top bearing coupling the top axle with the top of the enclosure.
(25) Other implementations of one or more of these aspects or other aspects include corresponding systems, apparatus, and computer programs, configured to perform the various actions and/or store various data described in association with these aspects. These and other implementations, such as various data structures for controlling the mechanical energy storage unit, may be encoded on tangible computer storage devices. Numerous additional features may, in some cases, be included in these and various other implementations, as discussed throughout this disclosure. It should be understood that the language used in the present disclosure has been principally selected for readability and instructional purposes, and not to limit the scope of the subject matter disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) This disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings in which like reference numerals are used to refer to similar elements.
(2) FIG. **1** depicts an example energy as a service platform.
(3) FIG. **2** depicts a block diagram showing example components of, and the interaction between, the utility server, a node, and an energy as a service manager.
(4) FIGS. **3**A-**3**D illustrate example mechanical-energy storage units from various angles and views.
(5) FIG. **4**A illustrates an example flywheel enclosure with a motor mount, magnetic coupling, and other features coupled therewith.
(6) FIGS. **4**B and **4**C illustrate an example enclosure tub.
(7) FIG. **5**A illustrates an example enclosure lid with various components attached thereto.
(8) FIG. **5**B illustrates an example vacuum assembly.
(9) FIGS. **6**A-**6**E illustrate various example flywheels and views thereof.
(10) FIG. **6**F illustrates example flywheel axles.
(11) FIG. **6**G illustrates cross section of an example bushing that applies pressure from a clamping plate on a stacking flywheel plate(s).
(12) FIG. **7**A illustrates a cross-sectional view of an example upper axle disposed within an upper bearing assembly of a flywheel enclosure.
(13) FIG. **7**B illustrates a cross-sectional view of an example lower axle disposed within a lower bearing assembly of a flywheel enclosure.
(14) FIG. **7**C illustrates an example flywheel coupled with a portion of a magnetic lift member and a lower bearing assembly outside of a flywheel enclosure.
(15) FIG. **8**A illustrates a cross sectional view of an example flywheel positioning system.
(16) FIG. **8**B illustrates a bottom-up view of an example nut bearing holder coupled with a flywheel enclosure.
(17) FIG. **8**C illustrates a top view of an example retaining cap.
(18) FIG. **8**D illustrates a top view of an example nut bearing holder held by a hex lock and retaining cap.
(19) FIG. **9**A illustrates an example assembly for an upper bearing of a flywheel assembly.
(20) FIG. **9**B illustrates a bottom view of example internal components of a magnetic lift member.
(21) FIG. **9**C illustrates a top-down view of an example magnetic lift member and bearing holder.
(22) FIG. **9**D illustrates example components of an example flywheel-motor coupling.
DETAILED DESCRIPTION
(23) This description includes several improvements over previous solutions, such as those described in reference to the Background. A mechanical-energy storage unit is described herein along with its assembly and an assembly fixture. Some aspects of the technology include components that reduce vertical force by a massive flywheel on bearings, such as an improved magnetic lift system. Other aspects include bearings, magnetic couplings, clamping plates, and/or other systems that may further improve the mechanical energy storage unit.
(24) In some implementations, one or two mechanical-energy storage units **150** may be installed at a residence to provide backup power in case of a power outage, to store electricity generated using residential solar panels, or to offset unevenness of power production and usage (e.g., an electrical utility may control the mechanical-energy storage unit **150** at a residence to address the balance energy use/production at the residence, nearby residences, or across the power grid). A mechanical-energy storage unit **150** may be buried next to an electrical panel or placed in a shed outside a residence, placed in a garage or utility room, or stored offsite.
(25) In some implementations, multiple mechanical-energy storage units **150** may be coupled together to scale energy backup at a larger facility, such as a business, or by an electrical utility. For instance, many mechanical-energy storage units **150** may be placed at a facility, buried, or otherwise used by an electrical utility. The multiple mechanical-energy storage units **150** may be communicatively linked to each other or to a central server to control storage and distribution of the stored energy (e.g., by controlling the rotational frequency of a flywheel **152** to keep various flywheels **152** at efficient speeds).

(26) Various implementations and features of flywheel energy storage systems (FESS) are described herein. These provide improvements over previous energy storage units including other flywheels **152**. For instance, the technology described herein provides an improved flywheel system or assembly, improved bearings, improved flywheel-motor couplings, improved flywheel housing, improved flywheel plates, improved assembly fixture, and method for assembly and use, among other improvements, features, and benefits.
(27) For example, a flywheel **152** may include a rotatable mass component, which may comprise a plurality of stacking plates **322**, cylinders, or other components, one or more bolt or clamping plates **320**, one or more axle **608** members, and other features. For instance, the technologies described herein include a plurality of plates that may have contoured edges based on an associated support structure, which allows increased speeds while reducing failure modes. For instance, the support structure may include clamping plates **320** that apply pressure to stacking plates **322**, thereby inducing friction between the plates to keep them in place and transfer rotational momentum between the plates and one or more axles **608**. In some implementations, two clamping plates **320** may be clamped together by bolts or other fasteners, which thereby cause the clamping plates **320** to apply pressure on massive plates (e.g., in an axial direction), which may be referred to herein as stacking plates **322**, and increase the friction among the stacking plates **322**, which may, in some cases, allow the stacking plates **322** to be used without other fasteners, thereby improving safety and efficiency. Other features and benefits of the flywheel **152** are described below. Not only are the plates improved, but their support structure is improved, among other benefits. Further implementations and features allow the expansion, positioning, and use of the flywheel **152** thereby further improving its performance.
(28) Among other improvements, the technologies described herein also include an improved support structure, such as an enclosure **304**, and support system, which may include, among other things, a sealed enclosure **304**, a lid-mounted vacuum assembly **308**, a magnetic coupling **318**, various bearings, and positioning mechanisms. The enclosure **304** may include a magnetic lift assist mechanism **352** that either entirely supports or partially supports the weight of the flywheel **152** (e.g., to reduce wear on bearings). The enclosure **304** may also include a transport surface and a lifting and adjustment mechanism that moves the position of the flywheel **152** internal to the enclosure **304** from a transport or storage position and adjusts it in an active position. The enclosure **304** may provide support for various components, such as a supercapacitor **306**, vacuum assembly, processor/controller/central processing unit, a motor **310**, and other components. The enclosure **304** may include various features for maintaining a vacuum, holding one or more bearings, positioning a flywheel **152** during use or transport, mitigating damage due to structural failures, and isolating vibration, among other features.
(29) Other benefits and features are described throughout this disclosure, but it should be noted that other features and benefits are contemplated. Furthermore, while various implementations are described in reference to the figures, these are provided by way of example and their features may be expanded, modified, or removed. For instance, features described in reference to some implementations may additionally or be used with other implementations.
(30) With reference to the figures, reference numbers may be used to refer to components found in any of the figures, regardless of whether those reference numbers are shown in the figure being described. Further, where a reference number includes a letter referring to one of multiple similar components (e.g., component **000***a*, **000***b*, and **000***n*), the reference number may be used without the letter to refer to one or all of the similar components. Further, it should be noted that while various example features and implementations are described throughout this disclosure and the figures, these examples are not exhaustive of every contemplated implementation, feature or permutation. For instance, while a certain feature may be described in reference to a first implementation, the feature may be used with a second implementation or the features, operations, etc., may otherwise be exchanged between the implementations.
(31) The innovative technology disclosed in this document also provides novel advantages including the ability to integrate modern technology with conventional power infrastructure; enable rapid transition to renewable energy sources; use the power grid as a backup; store power locally in nodes and regionalized storage clusters of nodes; isolate and minimize the impact of power outages; whether caused by natural disasters, infrastructure failure, or other factors; provide affordable alternatives to expensive and environmentally unfriendly electrochemical batteries; provide consumers the option to be independent from carbon-based power sources; and decentralize electric power production.
(32) As depicted in FIG. **1**, the innovative energy technology described herein may comprise an energy as a service platform (EaaS platform) **100**. The EaaS platform **100** may include an EaaS manager **110**, third-party server(s) **116**, user application(s) **172** operable on computing devices accessible to and interactable by user(s) **170** of the EaaS platform **100** and configured to send or receive data to the EaaS manager **110**, regionalized storage clusters **160** comprised of one or more nodes **140**, and the power grid **130** that comprises one or more power facilities **132** that are connected to a power transmission infrastructure **134**.
(33) A node **140** may be comprised of a power consuming entity and at least one ESU (e.g., an ESU **150** is provided as an example). A node **140** may be an entity that either consumers power itself or is coupled to entities that consumer power. In FIG. **1**, a node **140** is depicted as a premises **142**, such as a residential home, but it should be understood that any entity that consumes power is applicable, such as one or more appliances, a commercial structure such as a warehouse or office building, an electronic device or system (whether configured to move or static), a transportation system and/or vehicle, a transportation charging system, a power supply, a power substation, a power substation backup, etc. A regionalized storage cluster includes two or more nodes **140** in a given geographical region. A storage cluster may provide power banking functionality, as discussed further herein. The elements of the node **140** including the ESU(s) **150**, the independent power system **147**, the power grid **130**, and/or any appliances and/or other entities, may be electrically coupled via an electrical system **154** including wiring, junctions, switches, plugs, breakers, transformers, inverters, controllers, and any other suitable electrical componentry.
(34) In the depicted example, a node **140** is equipped with or coupled to power generating technology, such as an independent power system **147** and/or the power grid **130**. The independent power system **147** may comprise power generating technology that is localized and that allows for independent power generation, such as renewable power generating technology. Non-limiting examples include a solar electric system **144** (comprising a solar array, controllers, inverters, etc.), a wind turbine system **146** (comprising turbine(s), controllers, inverters, etc.), and/or other energy sources **148**, such as hydropower, geothermal, nuclear, systems and their constituent components, etc. The power generating technology may additionally or alternatively be conventional carbon-based power generating technology such as the depicted power grid **130**, although for carbon negative or neutral implementation, a greener power generating technology may be preferred.
(35) The node **140** may include or be coupled to an energy storage unit that is capable of storing excess power that is produced by the power generating technology. In some implementations, the energy storage unit may comprise a mechanical energy storage unit (ESU) **150**. Although the ESU **150** is illustrated and described herein as including one or more flywheels **152**A, **152**B . . . **152**N (also simply referred to individually or collectively as **152**), they may alternatively or additionally include chemical batteries, capacitors, or other energy storage devices. The ESU's **150** may convert the electricity received from the power generating technology to kinetic energy by spinning up (increasing the spin rate) of the flywheels **152** and/or by using one or more inverters that convert between direct current and alternating current, depending on the implementation.
(36) Each battery or flywheel **152** or may be configured to store up to a certain maximum amount of energy. By way of non-limiting example, a motor **310** coupled to the flywheel **152** may be configured to spin the flywheel **152** up to between 15,000 rotations per minute (RPM) and 25,000 RPM, such that the flywheel **152** may store between 18 kilowatt hours (kWh) and 28 kWh of electricity. Combined, multiple flywheels **152** could store between 54 kWh and 84 kWh of power. During hours in which the power generation technology, such as the solar cells, produce less power than what is consumed by the electrical apparatuses (e.g., appliances) of the premises **142**, the motor **310** may be operated as a generator that converts the kinetic (mechanical) energy stored in the flywheel **152** to electricity, thereby pulling power from the flywheel **152** to meet the local power needs of the node **140** (e.g., power the electrical apparatuses of the premises **142**). In this example, advantageously the node **140** may use an

average of 15 kWh of power daily and the ESU **150** is capable of powering the node **140** fully for about 4-6 days should the local power generating cease to produce any power.

(37) In some implementations, a premises **142** or node **140** may include or be coupled (physically or communicatively) with a local weather station **143** that measures light, precipitation, wind speed and direction, barometric pressure, or other weather data. For instance, where the premises **142** includes a residence, a local weather station **143** may be placed on a roof or in a yard of the residence. The local weather station **143** may be communicatively coupled directly with a controller **254** or central nervous system of a node **140** or ESU **150** or may be coupled with an EaaS manager **110**, third-party server(s) **116**, or another device via the network **102**, for instance. Accordingly, weather and other context data may be used, as described below, to train one or more machine-learning models that predict power consumption and/or production. The weather data may also be input into a trained model in order to predict future behavior (e.g. power consumption or production), as described elsewhere herein.

(38) In another example, as discussed further elsewhere herein, a utility may be integrated with the EaaS manager **110** and its utility management application **122** signal the power management application **111** via the storage cluster APIs **112** that it is experiencing a surge in demand for power, and the power management application **111** may signal a node **140** or cluster of nodes **140** (e.g., storage cluster **160**) to spin off power from the flywheels **152** and provide the energy back to the grid through the transmission infrastructure **134**, which may be connected to the node(s) **140** through connection points (e.g., two or three phase electrical service drops or buried power lines connected to a service panel, which typically includes power meter(s)). Conversely, the utility may be producing excess power and may wish to bank/store the power. The utility management application **122** may signal the power management application **111** via the storage cluster APIs **112** that it needs to store a given amount of power, and the power management application **111** may in turn signal a node **140** or cluster(s) of node(s) **140**, such as one or more regionalized storage clusters **160** to inform them of the storage need, and node(s) **140** in those storage cluster(s) **160** that have excess capacity and are configured to receive power from the grid may receive the power through the transmission infrastructure **134** and store it as mechanical energy in the ESUs for later retrieval. The EaaS platform **100** may charge the utility for the power banking service, as discussed further elsewhere herein.

(39) It should be understood that the RPMs and kWh figures provided in the prior paragraph are meant as non-limiting examples and that the ESU's **150** may be configured with flywheels **152** that are capable of storing more or less power depending on the implementation. For example, the weight of the flywheels **152**, the materials used for the flywheels **152**, the size and configuration of the flywheels **152**, the efficiency of the motor **310** and bearings, and so forth, may all be adjusted based on the use case to provide a desired about of back-up power for the node **140**. By way of further example, a flywheel **152** may be made of steel, aluminum, carbon fiber, titanium, any suitable alloy, and/or any other material that is capable of handling the cycles, vibration, radial and sheer stress and strain, and other conditions to which such a flywheel **152** would be subjected.

(40) The power transmission infrastructure **134** comprises a power network that couples power-consuming entities, such as homes, offices, appliances, etc., to power facilities that generate power from carbon, nuclear, and/or natural sources. The transmission infrastructure **134** may include intervening elements, such as step-up transformers, substations, transmission lines, and so forth, which are interconnected to provide power widely to different geographical regions.

(41) A power utility (also simply referred to as a utility), which may own and operate one or more power facilities and portions of the transmission infrastructure **134**, may operate a utility server configured to execute a utility management application **122**. The utility management application **122** may perform various functions such as load balancing, load managing, and grid energy storage, to manage the supply of electricity based on real-time demand. However, given the limitations of existing grid technologies, power outages, brownouts, and expensive peak power costs are still the norm.

(42) A user may use an instance of a user application **172** executing on a computing device, such as the user's mobile phone or personal computer, to configure and interact with the ESU(s) **150** that they are authorized to control, such as an ESU **150** installed at their home or business, as discussed further elsewhere herein.

(43) As shown in FIG. **1**, the EaaS manager **110**, the utility server **120**, the power facilities **132**, elements of the power grid **130**, the wind turbine system **146**, the solar electric system **144**, the other sources **148**, the node(s) **140**, the regionalized storage clusters **160**, the user applications **172** and associated computing devices, etc., may be coupled for communication and connected to the network **102** via wireless or wired connections (using network interfaces associated with the computing devices of the foregoing elements). The network **102** may include any number of networks and/or network types. For example, the network **150** may include one or more local area networks (LANs), wide area networks (WANs) (e.g., the Internet), virtual private networks (VPNs), wireless wide area network (WWANs), WiMAX® networks, personal area networks (PANs) (e.g., Bluetooth® communication networks), various combinations thereof, etc. These private and/or public networks may have any number of configurations and/or topologies, and data may be transmitted via the networks using a variety of different communication protocols including, for example, various Internet layer, transport layer, or application layer protocols. For example, data may be transmitted via the networks using TCP/IP, UDP, TCP, HTTP, HTTPS, DASH, RTSP, RTP, RTCP, VOIP, FTP, WS, WAP, SMS, MMS, XMS, IMAP, SMTP, POP, WebDAV, or other known protocols.

(44) The EaaS manager **110**, the third-party server(s) **116**, the utility server **120**, the node(s) **140**, the power facilities, and the user devices may have computer processors, memory, and other elements providing them with non-transitory data processing, storing, and communication capabilities. For example, each of the foregoing elements may include one or more hardware servers, server arrays, storage devices, network interfaces, and/or other computing elements, etc. In some implementations, one or more of the foregoing elements may include one or more virtual servers, which operate in a host server environment. Other variations are also possible and contemplated.

(45) It should be understood that the EaaS platform **100** illustrated in FIG. **1** and the diagram illustrated in FIG. **2** are representative of example systems and that a variety of different system environments and configurations are contemplated and are within the scope of the present disclosure. For example, various acts and/or functionality may be moved between entities (e.g., from a server to a client, or vice versa, between servers, data may be consolidated into a single data store or further segmented into additional data stores, and some implementations may include additional or fewer computing devices, services, and/or networks, and may implement various functionality client or server-side. Further, various entities of the system may be integrated into a single computing device or system or divided into additional computing devices or systems, etc., without departing from the scope of this disclosure.

(46) The third-party server(s) **116** one or more servers or services that provide additional or outside information. For example, a third-party server **116** may associated with and/or provide interaction with the National Weather Service™, an electric vehicle, a smart EV charger, a smart thermostat, a home automation system, a smart power meter (e.g., an Acrel™ meter), a solar power production service, or various other servers or services. For instance, the EaaS manager **110** and/or a node **140** (e.g., a controller **254** or central nervous system thereof) may communicate with one or more third-party servers **116** to receive power consumption data, power production data, contextual data, or otherwise. In some instances, the one or more third-party servers **116** may additionally or alternatively allow control of various power production devices or power loads.

(47) FIG. **2** depicts a block diagram showing example components of, and the interaction between, the utility server **120**, the node(s) **140**, and the EaaS manager **110**. The utility server includes an instance of the utility management application **122** and an EaaS interface **124**. The EaaS manager **110** includes a power management application **111**, utility APIs **112**, node **140** APIs **113**, and a data store **210**. The data store **210** may store and provide access to data related to the EaaS platform **100**, such as cluster data **211**, node **140** data **212**, user data **213**, usage data **214**, utility data **214**, analytics data **216**, and fault data **217**.

(48) In some implementations, the EaaS manager **110** may include a training engine **114** that receives various data and trains one or more machine-

learning models that may be used by a node **140**, EaaS manager **110**, or utility server **120** to estimate future behavior, such as consumption or production of power, as described in further detail below. It should also be noted that, although the training engine **114** is illustrated as being included on the EaaS manager **110**, it may be executed on a node **140**, utility server **120**, third-party server **116**, another device, or distributed among multiple devices.
(49) A node **140** of the EaaS platform **100** may include one or more ESU(s) **150**, although chemical or other forms of electrical storage are possible and contemplated herein. An ESU **150** may include an instance of a controller **254**. The controller **254** may include a flywheel coupler **255**, a flywheel selector **256**, and flywheel monitor **257**. The MESU (mechanical energy storage unit) hardware **258** may comprise a chassis, one or more flywheels **152**, magnets and/or bearings, a flywheel coupler **255**, and/or a motor-generator **310**. The motor-generator **310** may be coupled to each flywheel **152** via a flywheel coupler **255**. The flywheel coupler **255** may engage and disengage the motor-generator **310** from the flywheel **152**, such that each flywheel **152** may spin freely when disengaged and may be coupled to the motor-generator **310** when engaged such that the motor-generator **310** may increase the speed of the flywheel **152** (spin up the flywheel **152**), or the flywheel **152** may spin the generator to produce power. Each flywheel **152** may be levitated using magnets to minimize the friction caused by the rotation of flywheel **152**. As an example, a maglev unit may be used to suspend and retain the flywheel **152** while spinning.
(50) In some implementations, the ESU **150** may additionally or alternatively include battery hardware **259**, for example, for various other energy storage unit types, such as a chemical battery. For instance, one or more chemical battery cells, controllers, housings, or other configurations may be used.
(51) Additionally, or alternatively, bearings, such as but not limited to ceramic bearings, may be used to support and retain the flywheel **152** while spinning. The chassis may house and support the flywheels **152**. Flywheels **152** may be arranged horizontally or vertically. In horizontal orientation, flywheels **152** may have a wheel-like shape and may be stackable one on another in the same chassis, but still configured to spin independently of one another. In such a configuration, the coupler couple to each flywheel **152** independently, or more than one coupler and motor **310** may be used, depending on the implementation. In a vertical orientation, the flywheels **152** may have a roll like shape, and may be positioned parallel to one another in the chassis. In either orientation, in some implementations, the chassis may include a housing that encloses the ESU **150** and provides a vacuum environment in which the components of the ESU **150** may operate. This is advantageous as it may seal out dirt, debris, and corrosion causing elements, and allow for the flywheels **152** and other components to optimally operate.
(52) In some implementations, a node **140** may include one or more ESU(s) **150** and may act as a manager of the ESU(s), may receive and process information from the EaaS manager **110** for the two or more ESU(s) **150**, and may send signals to the ESU(s) **150** (e.g., via the controller **254**, MESU hardware **258**, and/or battery hardware **259**) and receive and process signals from the ESU(s) **150** (e.g., via the controller **254**, MESU hardware **258**, and/or battery hardware **259**), to control the functionality and operations of the ESU(s) **150**. In some further implementations, the structure, acts, and/or functionality of the controller **254** and the node **140** application **240** and their constituent components may be combined, and the node **140** may represent an ESU(s) **150** itself, to which one or more appliances that consume power may be coupled to receive power. Other variations are also possible and contemplated.
(53) The utility management application **122**, the controller **254**, the node **140** application, the node **140** management application, the utility APIs, the node **140** APIs, and the ESU APIs may each include hardware and/or software executable to provide the acts and functionality disclosed herein.
(54) Specifically, the utility management application **122** may be executable by the utility server **120** to monitor power generation and distribution by one or more power facilities and a transmission infrastructure **134**. The utility management application **122** may receive signals from various entities of the EaaS platform, such as the EaaS manager **110**, the nodes **140**, the transmission infrastructure **134**, other utility management applications associated with other providers, and so forth. The utility management application **122** may communicate via the EaaS interface **124** with the EaaS manager **110** to access the services provided by the EaaS manager **110**. In particular, the EaaS interface **124** may interact with the utility APIs of the EaaS manager **110** to request power banking, request the provision of supplemental power, to provide usage, performance, and/or demand data, and so forth. In some implementations, the EaaS interface **124** may generate and transmit a secure request via the network to the utility APIs. The power management application **111** may receive the request via the utility APIs and process it accordingly.
(55) In some implementations, a node **140** or premises **142** may include one or more local sensors, power meters, weather stations, and/or controllable devices, which, for example, may be electrically and/or communicatively coupled with the ESU **150** or other components of the node **140**. For instance, these sensors and devices may communicate directly with the controller **254** or with an EaaS manager **110** (directly or via one or more servers, networks, APIs, etc.). Accordingly, in some implementations, the training engine **114** may receive data from the devices **247** or other components of the node **140** (and/or other sources, such as third-party server(s) **116**) to train models. The training engine **114** may store the models in the node **140** data **212**, for example, if they are trained for a specific node **140**, cluster data **211** if they were trained for a cluster of nodes **140**, etc. Accordingly, the power management application **111** may receive or access the models to generate predictions, as described below, for example.
(56) The power management application **111** may be coupled to the data store **210** to store and retrieve data. The data stored by the data store **210** may be organized and queried using any type of data stored in the data store (e.g., cluster ID, user ID, utility ID, node **140** ID, ESU ID, configuration data, etc.). The data store **210** may include file systems, databases, data tables, documents, or other organized collections of data.
(57) Cluster data **211** may comprise information about a cluster of two or more nodes **140** (e.g., regionalized storage cluster), such as the identity of the node(s) **140**, the storage capacity of the storage cluster **160**, the availability of the storage cluster **160**, the operational health of the storage cluster **160** and/or constituent ESU(s) **150**, the historical performance of the storage cluster **160**, etc.). The cluster data **211** may include various consumption data, production data, weather data, context data, algorithms, equations, cluster attributes, machine learning models trained for the cluster, or other data.
(58) Node **140** data **212** may comprise information about a node **140**, such as the number of ESUs installed at the node **140**, the type of node **140**, the operational health of the ESU(s) **150**, any restrictions or operation parameters of the ESU(s) **150**, configuration data for the ESU(s) **150**, identifiers of the ESU(s) **150**, who the ESU(s) **150** belong to, whether the ESU(s) **150** can be used for banking grid power, whether the ESU(s) **150** have been inactivated, and so forth.
(59) The node **140** data **212** may include various consumption data, production data, weather data, context data, algorithms, equations, node **140** attributes, machine learning models trained for the node **140**, or other data, for example, as described in further detail elsewhere herein.
(60) User data **213** may comprise information about the user associated with a storage cluster **160**, node **140** or ESU **150**, including user account information, login information, user preferences governing the ESUs (e.g., schedule data, activation/inactivation data, etc.).
(61) Usage data **214** may comprise information about the usage of the clusters and/or ESU(s) **150**, such as spin rates of the flywheels **152**, power output levels, maintenance periods, downtime, inactive periods, third-party use (e.g., use by utilities or neighboring nodes **140**), etc.
(62) Utility data **215** may comprise information about utilities that have partnered with the EaaS platform **100**, such as utility account information, utility capability information, power banking requirements, contractual parameters, performance requirements, power grid **130** specifications, etc.
(63) Analytics data **216** may comprise insights about the EaaS platform **100**, such as local vs. grid power generation, aggregate usage data, aggregate performance data, etc. It should be understood that any other data that would be suitable and applicable to the EaaS platform **100** may be stored and processed by the EaaS manager **110**. In some instances, the analytics data **216** may describe performance of machine learning models, predictions, or other data, as described in further detail elsewhere herein.

(64) Fault data **217** may comprise sensor and other data, such as is described throughout this disclosure, including faults, alerts, thresholds against which data is compared to detect faults, and other data. For instance, fault data **217** may include any of the data described below in reference to the methods and graphical user interfaces, such as sensor data, voltages, state of charge, etc.
(65) The node **140** application **240** and the controller **254** may communicate with the EaaS manager **110** via the EaaS interface **242**, which is configured to interact with the node **140** APIs **113** surfaced by the EaaS manager **110**. The node **140** APIs **113** provide methods for accessing data relevant to the node **140** and the ESU(s) **150** associated with the node **140**, and executing various functionality, such as signaling unavailability/availability for banking power, requesting a functional upgrade, such as a higher spin rate for one or more flywheels **152** or deactivating/activating a previously active/inactive flywheel **152**, reporting usage data and/or state information, and so forth.
(66) In some implementations, the node **140** application **240** may executed on or in communication with a controller **254**, which may be a computing device that controls components of a node **140**, such as a mechanical battery (e.g., a flywheel), chemical battery, or various other components. For instance, the controller **254** may include a central nervous system (CNS) or computer that receives sensor data, provides instructions to inverters, motors, or other components, or performs other operations. For instance, the node **140** application **240** may be executed on the controller **254** or CNS to communicate with node **140** components and/or an EaaS manager **110** or another external component.
(67) The flywheel manager **244** of the node application **240** may be configured to communicate with the controller **254** to provide operational control signals, such as power banking signals, power extraction signals, spin rate adjustment signals, flywheel enablement/disablement signals, and so forth. The energy manager **246** is configured to communicate with the flywheel manager **244** and provides controls signals to the flywheel manager depending on the energy requirements (produce energy for local use, produce energy for utility use, bank local energy, bank utility-provided energy, etc.).
(68) The flywheel coupler **255** may be configured to control the mechanical coupling of the flywheels **152** with the motor-generator **310**, the flywheel selector **256** is configured to select which flywheels **152** to control based on the received control signals, and the flywheel monitor **257** monitors the state of each flywheel **152** for safe operation and performance within defined operational parameters, and can take control of the functionality of the MESU hardware **258** and shut down, slow, suspend, adjust, optimize, or other control the MESU hardware **258** depending on the monitored state.
(69) FIGS. **3**A-**3**D illustrate an example mechanical-energy storage unit **150** (MESU **150**) or flywheel assembly **150** from various angles and views. For example, FIG. **3**A illustrates a front-top view, FIG. **3**B illustrates a cross section view, FIG. **3**C illustrates a front-top view of another example MESU **150** or flywheel assembly **150**, and FIG. **3**D illustrates a front-top view with an enclosure lid **328** removed to show a flywheel **152** inside a flywheel enclosure **304** of an example MESU **150** assembly.
(70) The improved flywheel assembly **150** may be a mechanical-energy storage unit **150** with configurations and features that improve manufacturability while also providing redundancy, safety, and reliability that allow the flywheel assembly **150** to provide years of safe and relatively maintenance free operation in ways that were not previously possible, for example, as may be noted in the Background and elsewhere herein. Although various configurations are possible and contemplated, the illustrated example flywheel assembly **150** may include a vacuum enclosure **304** and support structure, a massive rotating flywheel **152** (not visible in FIG. **3**A) internal to the enclosure **304**, a motor-generator **310** that may be fully or partially external to the enclosure **304** and coupled with the flywheel **152**, a supercapacitor **306**, driver(s) and/or CPUs, inverter(s), circuit breakers, magnetic lift member(s) **352** (also referred to as magnetic lift components or mechanisms herein), bearings, physical or magnetic couplings **318**, a vacuum pump **308**, and various other components, as described below, although other implementations are possible and contemplated herein.
(71) The enclosure **304** is an example of a support structure that supports various components of a flywheel assembly **150**. While example enclosures **304** are shown completely enclosed, in some implementations, an enclosure or other support structure may be open, such as a cage, frame, or other configuration.
(72) The example flywheel assembly **150** may include, among other things, an enclosure **304** that is scalable to provide a vacuum, support to the flywheel **152** (not visible in FIG. **3**A), a mounting structure for various components of the assembly, and protection against mechanical failure, among other things. The example enclosure **304** for the flywheel **152** may be configured as a vacuum assembly case with reinforcement including features for coupling the flywheel **152** with the case, an improved shape, and an ability to adjust the flywheel **152** through the case. In some instances, a connection for creating and/or maintaining a vacuum may also be included with the case. Example implementations and features of the enclosure **304** may be described elsewhere herein, although other implementations are possible and contemplated.
(73) The flywheel enclosure **304** may be mounted and/or isolated from a mounting structure by one or more feet **314** or legs that couple the enclosure **304** with an external structure while isolating vibrations, as noted below, and may include mounting structures for accommodating various components of the assembly. For example, a supercapacitor **306** may be mounted to a support structure of or attached to the enclosure **304**, which supercapacitor **306** may buffer energy entering/exiting the flywheel **152**, for instance, by assisting the motor **310** to spin the flywheel **152** up or receive energy therefrom.
(74) Also, as described below, the enclosure **304** may provide mounting points or structures (e.g., a mounting brace **332**) for mounting a motor-generator **310** in line with the axis of rotation of the flywheel **152**, although, in other implementations, gears may be used to couple the motor-generator **310** with the flywheel **152**. The motor-generator **310** may be coupled with the flywheel **152** via one or more axle **608** components and, in some instances, a magnetic coupling **318** that allows a flywheel axle(s) **608** to remain physically decoupled from a motor-generator **310** rotor while still providing force to pass between them, although a physical coupling may also or alternatively be used, as described below. The motor-generator **310** may have an electrical connection to a supercapacitor **306**, inverter, driver, CPU, external grid connection or otherwise, which allows electrical current to flow into the motor-generator **310** to spin up the flywheel **152** or out of the motor-generator **310** to receive stored potential energy from the flywheel **152**.
(75) The motor-generator **310** may have various configurations, as noted in further detail elsewhere herein. In some implementations, the motor-generator **310** may be an electrical-vehicle motor or other motor (e.g., a Hyper 9™ motor), such as a brushless alternating current motor (e.g., a 3 phase AC synchronous reluctance internal permanent magnet motor) that can free-wheel in order to allow the flywheel **152** to store power for a longer period of time. The motor size and configuration may vary depending on peak output/input and flywheel **152** size/speed requirements. For instance, a smaller, residential MESU **150** may include a smaller size flywheel **152** with a 30-40 KW motor while a larger, commercial (e.g., for a store, electrical utility, subdivision, etc.) may have a 300-500 KW motor, although other implementations are possible. The motor controller and/or CPU may be the same for various sizes of flywheels **152** or may vary depending on the implementation.
(76) The motor-generator **310** may be coupled with the flywheel **152** using an axle **608** and bearing of the flywheel **152**/flywheel enclosure **304**. Similarly, in some implementations, the flywheel axle **608** and motor-generator **310** may be coupled using a flywheel motor coupling, which may include a direct connection, magnetic coupling **318**, friction clutch, torque converter, gearbox, or otherwise, as described in further detail below.
(77) An example flywheel **152** (not visible in FIG. **3**A) may be housed in and/or supported by the enclosure **304** and components thereof. Example flywheels **152** and features thereof are described throughout this disclosure. For example, a flywheel **152** may include a plurality of stacking plates **322** held together by a support structure, such as clamping plates **320** (e.g., using compression and friction). The support structure may include one or more axles **608** that attach thereto and provide support to the flywheel **152**. As described in further detail elsewhere herein, the configuration of the support structure and axles **608** may allow flywheel plates to be used without the axle **608** perforating the plates. Depending on the implementation, the axle(s) **608** may be vertically and/or horizontally supported by other components or the flywheel enclosure **304** and may

couple with a motor-generator **310** (e.g., as noted above).
(78) For example, an axle **608** may interact with one or more bearings, whether magnetic, metal, ceramic, hybrid ceramic, etc., of the enclosure **304**, to allow the flywheel **152** to spin about an axis formed by the axle **608**. The enclosure **304** may include or couple with one or more bearings that support the flywheel **152** horizontally to keep it spinning with little-to-no vibration, as described below. In some cases, the bearings may be ceramic to avoid interaction with a magnetic field of a magnetic lift member **352** or other magnetic component.
(79) The axle(s) **608** may interact with the bearing(s) to provide vertical or horizontal support to the flywheel **152**, for example, by keeping the axles at a defined location and balanced at a defined axis of rotation.
(80) For instance, a bottom bearing may interact with the bottom of the flywheel **152** and/or a top bearing may interact with the top of the flywheel **152**, for example, inside the enclosure **304**. In some implementations, a magnetic levitation device or magnetic lift assistance member **352** may be used to reduce the friction or pressure, for example, on one or more of the bearings. For instance, a magnetic levitation device may be disposed at a bottom of the flywheel **152** to apply upward force thereon thereby limiting the force due to gravity on a bottom bearing and/or balancing force between a top and bottom bearing. In some implementations, a magnetic lift assistance member (also referred to as magnetic lift member) **352** may be positioned at a top of the flywheel **152**/enclosure **304** to pull the enclosure **304** upward, thereby decreasing the force due to gravity on a bottom bearing(s). As noted in further detail elsewhere herein, the magnetic lift assistance member **352** may lift less then, exactly, or greater than the weight of the flywheel **152** so that there is some, little, or no weight on the bottom and/or top bearing(s).
(81) The amount of weight held by magnets of the magnetic lift assistance member **352** may be adjusted based on a distance from the magnets, as described elsewhere herein. For instance, the flywheel **152** may be manually adjusted by an installer or, in some implementations, the enclosure **304** may include or may be coupled with one or more flywheel positioning components that may adjust the position of the flywheel **152**, for example, to ensure that a correct distance between the flywheel **152** and magnetic lift assist mechanism **352**, top bearing, bottom bearing, or other component of the assembly **150**. For instance, a flywheel positioning component may move the flywheel **152** (e.g., inside the enclosure **304**) from a shipping position to an engaged position where it is in a correct position relative to the bearing(s) to minimize bearing wear and friction.
(82) It should be noted that although the enclosure **304** is illustrated as fully enclosed, including reinforcements, welds, seals/O-rings, etc., that allow a vacuum to be maintained inside the enclosure **304** with the flywheel **152**; however, it should be noted that other implementations are possible and contemplated herein, such as where the enclosure **304** is fully or partially open.
(83) In some implementations, the flywheel assembly **150** may include various components mounted to the enclosure **304** (e.g., via a lid **328** assembly mounting plate or bracket) that support the operation of the flywheel **152**. For instance, the flywheel assembly **150** may include a supercapacitor **306**, motor-generator **310** (and associated mounting hardware), driver and CPU/controller **312**, vacuum pump **308**, various inverters, wiring harnesses, circuit breakers, and other equipment, although other implementations are possible and contemplated herein.
(84) As illustrated in the examples of FIGS. **3**A-**3**D, a flywheel enclosure **304** may be round with a flat bottom and top and various reinforcing ridges, which configuration may provide strength to the enclosure **304** to prevent buckling due to an internal vacuum while also preventing external damage in case of a mechanical failure of the flywheel **152**. It should be noted that the enclosure **304** may be square, hexagonal, etc. It may have rounded (e.g., as illustrated in FIG. **3**D) or flat sides (e.g., as illustrated in FIG. **3**A). As illustrated in FIG. **3**A, a mounting plate may be positioned on top of the enclosure **304** components mounted thereto, as described in further detail below.
(85) In some implementations, as illustrated in the examples, the flywheel assembly **150** may include one or more (e.g., **3** and **4** arms are illustrated) motor mount braces **332** that couple with a lid **328** of the flywheel enclosure **304** and extend upward to support a motor mount, which may comprise a ring that holds the motor-generator **310** in alignment with an axis of rotation of the flywheel **152**. In some instances, the motor mount braces **332** may include linear actuators that lift the motor-generator **310** vertically in order to decouple the motor-flywheel **152** coupling, such as the magnetic coupling **318** described in further detail below. The flywheel assembly **150** may include a component mounting plate or lid assembly mounting plate(s) **330** that couple with top ribs or other structures of the enclosure **304** and provide mounting points for the various accessory components of the flywheel assembly **150**. In some instances, the lid **328** or lid assembly mounting plate(s) **330** may have various perforations that allow the motor mount braces **332**, axles **608**, motor-flywheel **152** coupling, vacuum pump **308** connection and other components to pass therethrough. Accordingly, the components may be securely mounted to mounting plate **330**, lid **328**, enclosure **304**, or otherwise (e.g., as illustrated in the example figure) in order to speed assembly and improve stability.
(86) As shown in the example of FIG. **3**A, the enclosure **304** may have a plurality of reinforcing structures, such as ribs, rings, etc.
(87) The enclosure **304** may also include one or more feet **314** or other supports that provide support to the ribs (e.g., the side or bottom ribs) or other structures (e.g., a bottom plate) of the enclosure **304** to secure the flywheel assembly **150**, support the weight of the flywheel **152**, and/or isolate the flywheel **152**'s movement/vibration; although, it should be noted that vibration is ideally limited by balancing the flywheel **152** and acceleration, temperature, or other sensors may be located in the bearings, axles **608**, enclosure **304**, or other components.
(88) FIG. **3**B illustrates an example cross section view of a flywheel assembly **150**. For instance, as shown in the figure, a flywheel **152** having a number of stacking plates **322** and a top and bottom axle **608***b* is located inside an enclosure **304**. The bottom axle **608***b* of the flywheel **152** is shown interacting with bottom bearings that support the flywheel **152** horizontally and/or vertically. The top axle **608***a* of the flywheel **152** is shown passing through a magnetic lift member **352** and into a magnetic coupling **318**, which couples the axle **608** with a stator (directly or via other components, axles **608**, drive shafts, gears, etc.) with the motor-generator **310**, which is held vertically above the axle **608** using the motor braces **332**. Additionally, as noted elsewhere herein, various sensors may be located throughout the assembly, such as the RPM sensor mount **316** that is located adjacent to the magnetic coupling **318**, as well as various temperature, acceleration, etc., sensors that may be positioned adjacent to the motor **310**, bearings, and other components of the assembly. These and other implementations and features are described in further detail below.
(89) FIG. **3**C illustrates another example MESU **150** or flywheel assembly **150** with a different implementation of the enclosure **304**. As illustrated, an enclosure **304** may be a cylindrical enclosure with an enclosure base, such as a tub **326** and a lid **328**. The enclosure **304** may also include one or more feet **314** (e.g., three are illustrated in FIG. **3**C) or legs support the flywheel assembly **150**. A foot **314** may include a bushing or other component that isolates vibrations, bolt holes to bolt the flywheel assembly **150** to a floor or other location.
(90) In the depicted example, the motor-generator **310** may be mounted higher on motor mount braces **332** and/or base than the example of FIG. **3**A, for example, to allow access to mount or remove the motor-generator **310**, magnetic coupling **318**, bearings, or other components. Additionally, while the other components illustrated in FIG. **3**A are not shown in FIG. **3**C, they may also be mounted to the lid **328** or another location of the flywheel assembly **150**. For instance, a vacuum pump **308**, supercapacitor **306**, chemical battery, driver, CPU, etc., may be mounted to the lid **328**, tub **326**, other portion of the flywheel assembly **150**, or otherwise.
(91) FIG. **3**D illustrates the example flywheel assembly **150** of FIG. **3**C with the lid **328** and other components omitted to show an example massive flywheel **152** inside the enclosure **304**. As shown, the flywheel **152** may be positioned at a center of the enclosure **304**, although other implementations are possible. As illustrated in the example of FIG. **3**D, a flywheel **152** may include one or more clamping plates **320** (the top clamping plate **320***a* is shown), one or more stacking plates **322**, one or more bolts holding the clamping plates **320** together and/or to the stacking plates **322**, and one or more axles **608**. These and other features and implementations are described in further detail elsewhere herein.
(92) FIGS. **4**A-**5**B illustrate an example flywheel enclosure **304** and various components, views, and constructions thereof. FIG. **4**A illustrates an example flywheel enclosure **304** with a motor mount, magnetic coupling **318**, and other features coupled therewith. For instance, various components, such as the motor-generator **310**, CPU, vacuum **308**, and mounting plates **330** are removed to expose the underlying structure. As

illustrated in the example, four motor mount braces **332** are coupled to a top plate (e.g., part of a lid **328**) of an enclosure **304** and extend upward to provide rigid support to a motor mount base **406**. The motor mount base **406** may be round or any other shape to accommodate various components, such as the motor-generator **310**, magnetic coupling **318**, bearing(s), axle(s) **608**, etc. For instance, the motor mount base **406** may be rounded at a top to receive and mount (e.g., using fasteners, such as bolts) the motor-generator **310**, have a perforation through which a flywheel axle **608** and/or motor axle, etc., may pass, and or may allow various other components to be coupled or mounted thereto. For instance, a magnetic coupling **318** may be mounted to or integrated with the motor mount base **406**, so that it can easily be attached to the enclosure **304**.

(93) As depicted in the example, the motor mount braces **332** and base **406** may be configured to be lifted above, accommodate, and/or hold other components. For instance, a motor coupling, such as the illustrated example magnetic coupling **318** may be coupled to a bottom side of the motor mount base **406** to interact with both a top axle **608*a*** and a motor-generator **310**. Similarly, this positioning may allow top bearing(s) to be installed or maintained under the braces/base. Similarly, as noted elsewhere herein, reinforcing components or structures of the flywheel assembly **150** may support the motor **310**, prevent undesired twisting of components, and hold a bearing and/or magnetic lift member **352**.

(94) FIG. **4**A also illustrates various assembled structures of the flywheel enclosure **304**, which may include a lid **328** and a tub **326**, for example. The lid **328**, as described in further detail below, may include a top plate, top rib(s), motor mount brace(s) **332**, mounting plates, and various other structures. For instance, a top ring **418** may include a ring of material (e.g., a steel ring or set of bends in steel plates) may be disposed surrounding an axle **608*a***/axis of rotation to provide torsional rigidity to the enclosure **304** and/or to top ribs, which may radiate outward from (e.g., merge at/into) the top ring **418** and provide strength to the lid **328**, which may support motor mount brace(s) **332**, mounting plate(s), a vacuum **308**, etc., and may prevent the top plate from buckling when force is placed thereon. The top ring, top rib(s), top plate, and/or other structures may work together to support air pressure due to an internal vacuum and/or the weight of the flywheel **152**. For instance, a magnetic lift mechanism **352** may be coupled to the lid **328** (e.g., at a center near the axle **608*a***) to lift some or all of the weight of the flywheel **152**, so the strength of the lid **328** is particularly beneficial. Depending on the implementation, the lid **328** may be constructed of one quarter to one half inch steel plate, or a thicker construction (e.g., as in the example of FIG. **3**C), which may be flat, welded together, and/or have various bends to further increase rigidity. For instance, the top ring **418** and top ribs may include one or more longitudinal bends to increase their strength and the ability to mount components thereto.

(95) The enclosure tub **326**, as described in further detail below, may include one or more side walls that encircle the flywheel **152**, which may be a continuous ring of material or bent metal (e.g., steel) or other plates that are welded together. The side walls **432** may provide vertical strength to the enclosure **304** while also mitigating mechanical failure of the flywheel **152**. Side ribs **434** (e.g., steel plates welded to the side walls **432**, such as the top ribs **524**) may also be attached around the side wall **432**, as illustrated, to provide further strength and avoid buckling inward or outward. The side ribs **434** and/or side walls **432** may be coupled (e.g., welded, glued, bolted, etc.) with a ring **430** to which a lid **328** may be bolted, as described below, and with a bottom plate **436** (which may have structures, such as bottom ribs **428**, as described below).

(96) For instance, a magnetic coupling **318** is shown in the example of FIG. **4**A. The magnetic coupling **318** may couple the flywheel axle **608** (e.g., **608*a***) with an axle or rotor of the motor-generator **310**. The magnetic coupling **318** may be supported by a motor mount base **406**, top ring **418**, or other components of the flywheel assembly **150** to hold it above an axle **608**. The magnetic coupling **318** may include an external rotor bottom **410** and external rotor top **412**, which may house an internal arrangement of magnets and/or bearings, etc., as described in further detail below. The magnetic coupling **318** may include an internal rotor top **414** with a rounded machine key **416** that interacts with a corresponding slot in an axle and/or rotor of the motor-generator **310** to improve the strength of the mechanical connection between these components. A similar structure may additionally or alternatively be used with an axle **608*a***. The magnetic coupling **318** is described in further detail below.

(97) FIG. **4**B illustrates an example enclosure tub **326** shown from the bottom. As illustrated, a lid **328** has been omitted while the bolts for the lid **328** are shown in place in a top ring **430**. As described in further detail above, the enclosure tub **326** may include side wall(s) **432**, side rib(s) **434**, a bottom plate **436**, bottom rib(s) **438**, feet **314**, and other components. For instance, a bottom reinforcement ring **440** may be located at a center of the bottom plate **436** and bottom ribs **438** may radiate outward therefrom in order to provide strength and rigidity. The ribs and other structure illustrated, as noted with the top **328**, may provide support to the flywheel **152**, for example, via bearings, magnetic lift, and/or axle(s) **608**.

(98) Four example feet **314** are illustrated coupled with the bottom plate **436** of the tub **326** in the example of FIG. **4**B. The feet **314** may allow the flywheel assembly **150** to be bolted directly to an external structure and, in some instances, may allow some leveling of the enclosure. Example implementations of feet **314** are described in further detail below.

(99) FIG. **4**B also illustrates a retaining cap **442** that may couple with the enclosure tub **326** (e.g., by bolting it to a bottom reinforcement ring **440** or other structure) to allow various components, such as a bottom/lower bearing assembly of the flywheel **152** to be accessed. The retaining cap **442** may also include seals/O-rings that seal the vacuum of the enclosure **304**. As described in further detail below, the bottom or lower bearing assembly may include one or more horizontal and/or vertical bearings, a shipment support area or ring **462**, a bearing height adjustment mechanism (e.g., the nut bearing holder **464** described below), and adjustment locking mechanism.

(100) For example, FIG. **4**B illustrates an example enclosure tub **326**. As illustrated, in the example implementations, various configurations and constructions are possible. For instance, the enclosure **304** and its components may be made of plate metal (e.g., steel, aluminum, etc.) that is attached together to form the enclosure **304**. For instance, the plate metal may be coupled or bent at various angles or continuously to create side walls **432**, a bottom plate **436**, top lid **328** (not shown in FIG. **4**B), etc., and ribs (e.g., **434**) positioned at a normal angle to the plates may be attached thereto to provide increased strength. The walls, ribs, top/bottom plates, and other components may be attached together using various techniques, such as welds, glue, or fasteners. For instance, where metal plates are used, they may be welded together to not only provide strength against an internal vacuum but also mitigate against mechanical failures of the flywheel **152**. For example, a bottom plate **436** may be welded to a side wall **432** (which may comprise one or multiple coupled segments), which may be welded to a top ring **430**, as shown in FIG. **4**C. Side **434** and bottom ribs **438** may be welded to the walls, plates, and/or reinforcing rings, as illustrated in the examples.

(101) Other implementations of an enclosure tub **326** or other components are described and illustrated elsewhere herein.

(102) In some implementations, the top ring **430** (and/or the lid **328**) may include grooves **452** for accepting one or more seals or O-rings, so that top ring **430** may be sealed against the lid **328**, although other implementations are possible. In some instances, multiple (e.g., two) seals/O-rings may be used to provide redundancy.

(103) In some implementations, a side wall **432** may include one or more holes or perforations through which the internal cavity may be accessed, such as for adjustment, sensors, for receiving a vacuum hose or fitting, or for other purposes. These perforations may be sealed using gaskets, caps, or other components during operation of the flywheel **152**.

(104) As illustrated in FIG. **4**C, in some implementations, the enclosure tub **326** may include a hole **454** or perforation at a bottom center at which the bottom/lower bearing assembly may be located. For instance, the hole **454** in the bottom of the tub **326** may allow the flywheel **152** to be adjusted, mounted, or otherwise, as described elsewhere herein. It should be noted that in other implementations, such as where a solid bottom of the enclosure **304** supports bearings and/or other components of the flywheel **152**, are possible. For instance, the enclosure may have a cylindrical shape, as illustrated herein, although other implementations are possible. Similarly, in some implementations, the flywheel **152** may be entirely supported by a lid **328** of the flywheel enclosure **304** and the bottom of the flywheel enclosure **304** may be solid.

(105) FIG. **4**C illustrates an example enclosure tub **326** with a shipping ring **462**, nut bearing holder **464**, and bearing(s) **466**, which may be mounted at the hole or perforation in the bottom of the enclosure **304**. As described in further detail below, one or more bearings **466** that support the flywheel **152** (e.g., via a bottom/lower axle **608*b*** of the flywheel **152**) may be held by a nut bearing holder **464**, which may be vertically

adjustable to move the bearings up or down. Accordingly, by adjusting the bearing holder **464**, the flywheel **152** can be moved up or down to move it between a storage position and adjust it in the enclosure **304** to provide an appropriate amount of force on the top and/or bottom bearings **466**.
(106) For example, a shipping ring **462** may be located at the bottom of the enclosure tub **326**, so that a portion of the bottom axle **608***b* (not shown in FIG. **4**C) and/or bottom clamping plate **320***b* (e.g., a flat bottom portion thereof, as illustrated in other figures herein) may rest thereon when in a shipping position. For instance, when a nut bearing holder **464** is adjusted into a shipping position (e.g., completely downwards), a bottom bearing **466** and/or top bearing may be fully disengaged (e.g., vertically) from the flywheel **152** to avoid damage to the bearings during shipping, for example, where the weight of the flywheel **152** is supported on the shipping ring **462**. As described in further detail below, the bottom bearing **466**/bearing holder **464** may be adjusted to move the flywheel **152** (e.g., vertically upward) to engage a top and/or bottom bearing. Additionally, or alternatively (e.g., where a top bearing has a variable position), the flywheel **152** position may be adjusted to vary a distance to a magnet of the magnetic lift member **352** using the nut bearing holder **464**, for instance.
(107) In some implementations, the nut bearing holder **464** may be un-adjustable, fixed, or omitted (e.g., replaced by another bearing holder). Additionally, or alternatively, a shipping ring **462** may be omitted or may be removable. For instance, a shipping ring **462** or other structure may be installed during assembly and then removed during installation, for example, as a flywheel **152** is manually adjusted by an installer.
(108) FIG. **4**C also illustrates an angled flange **468** in a side wall **432** of the enclosure **304**, which may allow access to an internal cavity, such as by a sensor, vacuum assembly **308**, or other components of the flywheel assembly **150**.
(109) In some implementations, a nut bearing holder **464** may be positioned within the base ring **440** and shipping ring **462**/bolt plate may move upwards and downward therein in order to move the bearing(s) **466**. The bearings **466** may move upward and/or downward with the nut bearing holder **464**, for example, to lift or lower the lower/bottom axle **608***b* and change the position of the flywheel **152**, as described above. A bottom axle **608***b* of the flywheel **152** may extend into the nut bearing holder **464** to interact with the bearings **466** held thereby.
(110) FIG. **4**C illustrates an example enclosure tub **326** with a shipping ring **462** and bearing(s) **466**, which may be mounted at the hole or perforation in the bottom of the enclosure **304**. In the depicted example, a nut bearing holder **464** may be omitted or permanent, for example, where it is not adjustable, although other implementations or combinations are possible. In some implementations (e.g., as in FIG. **3**C, a side wall **432** may be a circular wall, for example, with or without ribs. For instance, a thicker side wall **432** may be continuous or a strip welded at the ends to form the side wall **432**. The tub **326** may include a bottom plate **436** welded or integrated with the side wall **432**.
(111) FIG. **5**A illustrates an example enclosure lid **328** with various components attached thereto. For instance, the enclosure lid **328** may be placed onto an enclosure tub **326** (not shown in FIG. **5**A) to form an enclosure **304**, which may be vacuum sealed, depending on the implementation. As illustrated in the example, a lid **328** may include a top plate **522** with reinforcing top ribs **524** that extend radially from an axle **608***a* (e.g., from a top reinforcing ring **418**) to an outer edge of the plate **522**. In some instances, the top ribs **524** may extend beyond the top plate **522** or into cuts in the top plate **522**. For example, the top rib(s) **524** may extend partially (e.g., at an end) into slots formed in the top plate **522** to further enhance rigidity and case manufacturability.
(112) Other configurations of a lid **328** are also possible, such as the example implementation of FIG. **3**C. For example, a lid **328** may not have reinforcing ribs mounting plates (e.g., **330**) or other components, such as where the top plate **522** is thick enough to support the flywheel **152**, a magnetic lift component **352**, or other component. In some instances, the top plate **522** may include a recess or other area to accommodate, receive, or couple with the magnetic lift member **352**.
(113) The lid **328** may also include O-rings, O-ring grooves/channels **532**, or other seal locations around a periphery of the top plate **522**, center perforation (e.g., in association with a top bearing assembly or other components), and other features for sealing the enclosure **304** when the lid **328** is attached to the enclosure tub **326** (e.g., by bolts around the peripheral edge). In some instances, the lid **328** or other components may include a hole, seal, valve, etc., through which a vacuum assembly **308** may be attached in order to actively establish or maintain a vacuum. For example, as noted above, a vacuum assembly **308** may be mounted to a lid **328** assembly mounting plate **330** or otherwise, depending on the implementation.
(114) In some implementations, the lid **328** may also include a motor **310** mounted thereto, along with other components, such as a driver, controller/CPU **312**, supercapacitor **306**, etc. As these and other components may be previously assembled on the lid **328** and then placed onto the enclosure tub **326** (e.g., where a flywheel **152** is already positioned in the tub **326**), which may improve the speed and case of assembly.
(115) In some implementations, the lid **328** may include a perforation at an axis of rotation of the flywheel for receiving a top axle **608***a* of the flywheel **152**, although other implementations are possible, such as where a top axle **608***a* interacts with a magnetic coupling **318** integrated or coupled with the lid **328**. For instance, the magnetic coupling **318** may be sealed and/or placed at a center of the lid **328** and may interact with the top clamping plate **320***a* to provide interaction between the flywheel **152** and the motor **310**.
(116) In some implementations, the axle **608***a* may pass through the perforation, which may include or be coupled with one or more bearing(s) **334** that support the axle **608***a* horizontally and/or vertically (e.g., holding the flywheel downward from contacting magnets in the magnetic lift member **352**). One or more magnets, such as in a magnetic lift assist member/mechanism **352** may be attached to the lid **328**.
(117) A magnetic lift member **352** may extend downward from the bearing(s) **334** or other components to bring it into proximity with the top clamping plate **320***a* and/or stacking plates **322** of the flywheel **152**, which may increase the efficiency of the magnets. Although the magnetic lift member **352** is illustrated as being a continuous ring, multiple individual magnets may be included (e.g., in a balanced manner) around the axis of rotation of the flywheel **152** (e.g., inside a housing of the magnetic lift member **352**). In some implementations, the height of the magnetic lift member **352** and/or its magnets may be adjustable by tightening or loosening bolts coupling the magnetic lift member **352** to the lid **328**, for example, from underneath the lid **328** or on top of the lid **328** (e.g., when the lid **328** is on top of the enclosure tub **326**). Accordingly, a position (and, by extension, strength) of the magnets may be adjustable to further balance the system and force on the bearings **334**.
(118) FIG. **5**B illustrates an example vacuum assembly **308** that may actively maintain or initially established a vacuum in the enclosure **304** (e.g., via a perforation in the lid **328** or tub **326**). The vacuum assembly **308** may be mounted to the enclosure **304**, as illustrated in FIG. **3**A, and it may be triggered using a pressure sensor that senses pressure inside the enclosure **304**. The vacuum assembly **308** may include a vacuum pump **552** that is powered by the flywheel **152** itself, supercapacitor **306**, a chemical battery, or grid power. The vacuum pump **552** may be coupled with a motorized on/off valve **554** that opens or closes the vacuum to avoid leakage, a solenoid valve **556** and solenoid valve coil housing **558** that may allow air to enter the enclosure **304**, desiccant filter **560** that prevents dust or debris from entering the enclosure **304** while also reducing buildup of moisture (e.g., due to the operation of the vacuum pump **552**), and other components that maintain the vacuum and limit humidity in the system.
(119) The vacuum assembly **308** may include additional, fewer, or different components. It may be used to reduce a pressure and therefore an air resistance of a spinning flywheel **152**. In some implementations, when a technician is performing maintenance or repairs on the flywheel assembly **150**, the vacuum may be released (e.g., where air enters the enclosure **304** through a filter) to allow the maintenance to be performed. In some implementations, the vacuum pump **552** may create a positive pressure inside the flywheel enclosure **304**. By providing a positive pressure, dust or other debris may be prevented from entering the enclosure **304**, for example, because it may be difficult to clean out.
(120) FIGS. **6**A-**6**G illustrate an example flywheel **152** and various components, views, and constructions thereof. There are a number of innovative features in the flywheel **152**. For example, the flywheel **152** may include flywheel plates that are coupled together using friction, which may be performed in addition to or in lieu of other connections, such as adhesive, welding, or otherwise. Some implementations of the flywheel **152** include bolts through components while others do not include bolts through components. Similarly, some implementations of the flywheel **152** include two separate axles **608**-*a* top axle **608***a* and a bottom axle **608***b*. For instance, while previous flywheels **152** may include bolts attaching

each of their components together, some implementations of the flywheel **152** herein may separate the axle **608** and/or use a clamping force from clamping plates **320** (and/or axles **608**) to increase friction between the stacking plates **322** themselves, which may improve manufacturing and reduce points of failure when the flywheel **152** is spinning at high speeds.

(121) In some implementations, clamping plates **320** may be used on the top and bottom of the flywheel **152** to support the flywheel **152**, for example, by coupling the stacking flywheel plates together and/or to axles **608**. A top clamping plate **320***a* and a bottom clamping plate **320***b* may be drawn together by bolts at or near its peripheral edge, which applies pressure inward on the stacking plates **322** in an axial direction thereby increasing friction. The friction allows rotational force to be transferred through the stacking plates **322** while also preventing them from moving out of alignment, which may throw the balance of the flywheel **152** off.

(122) Depending on the implementation, the clamping force from the clamping plates **320** may be applied to the stacking plates **322** directly (e.g., by direct contact between the clamping plates **320** or stacking plates **322**) or via other components, such as a portion of an axle **608** or other contact points. For example, a clamping plate **320** may apply force to a center of the stacking plates **322** via a top and bottom axle **608***b* (and/or washer(s), bushings at a peripheral edge, or otherwise).

(123) In some implementations, the clamping plates **320** may be less massive than the stacking plates **322**, so each type of plate may expand (and, potentially, become thinner) differently, especially at the peripheral edge. Accordingly, in some instances, bushings or other components may allow the stacking plates **322** to move relative to the clamping plates **320** while the clamping force is continuously applied.

(124) The clamping plates **320** may have various contours and configurations to allow them to provide clamping force and other functionality. In some implementations, the stacking plates **322** may be configured differently from the clamping plates **320** and their function is primarily to add rotational mass to the flywheel **152** in order to store energy. The stacking plates **322** may be massive plates that are substantially round or may include various contours based on interaction with the clamping plates **320** or an assembly fixture, as described in further detail elsewhere herein.

(125) FIG. **6**A illustrates a side-top view of the example flywheel **152** and FIG. **6**B illustrates a side-bottom view of the example flywheel **152**. As illustrated, the stacking plates **322** may be continuously stacked with their faces touching each other to minimize space consumed and flex while increasing friction. Fourteen stacking plates **322** are illustrated, although other implementations are possible and contemplated herein. As illustrated, there may be a space between one or both of the clamping plates **320** and the stacking plates **322**. Although this space is illustrated as being relatively large and uniform, it may be smaller. For instance, there may be only a few millimeters between the bottom clamping plate **320***b* and the bottom-most stacking plate **322**, which space may vary based on clamping force applied and flex of the clamping plate **320**.

(126) As illustrated in the example flywheel **152** of FIG. **6**B, a bottom axle **608***b* may be coupled with a bottom clamping plate **320***b*. The bottom clamping plate **320***b* then interacts with a bottom flywheel stacking plate **322** (e.g., via bushings, an axle washer, a portion of the axle **608**, etc.). Various quantities of stacking flywheel plates may be stacked together depending on desired energy capacity, as noted elsewhere herein. Similarly, a top clamping plate **320***a* may interact with a top-most flywheel stacking plate **322** (e.g., via bushings, axle washer, etc.). The top clamping plate **320***a* may be coupled with a top axle **608***a*. In other implementations, a bottom face of the top clamping plate **320***a* may rest directly against the top face of the top-most stacking plate **322**.

(127) In some implementations, each of the stacking plates **322** may be identical, and each of the clamping plates **320** may be identical, although other implementations (e.g., sizes, configurations, etc.) are possible and contemplated, as noted below. Similarly, the top and bottom axle **608***b* may be the same or different (e.g., having a different length, interacting with different bearings or configurations, as illustrated herein.

(128) As illustrated, when assembled, the clamping plates **320** of the flywheel **152** may align with the stacking plates **322**. In some implementations, a clamping plate **320** may have a star shape (e.g., as illustrated in FIGS. **6**A and **6**B) where the tip of each arm or branch of the clamping plate **320** has a bolt hole that receives a bolt for clamping the clamping plates **320** together. In some implementations, a clamping plate **320** may have another shape (e.g., as illustrated in FIGS. **6**D and **6**E) including one or more perforations proximate to a peripheral edge.

(129) Similarly, the configuration of the stacking plates **322** may be based on the shape (e.g., the position and quantity of branches of the clamping plate **320**), as described in further detail below. For instance, bolt points of the stacking plates **322** may correspond to bolt points of the clamping plates **320** whether or not, as illustrated in the example of FIGS. **6**A-**6**C, the stacking plates **322** do not contact the bolts.

(130) Although not visible in FIGS. **6**A and **6**B, in some implementations, various mechanisms may be used at the axle-to-clamping-plate interface to keep the axle **608** and clamping plate **320** mechanically connected, so that rotational force may be transferred between them. For instance, the hole in the clamping plate **320** that accepts a portion of the axle **608** may have an oval shape or a flat area (e.g., to be shaped like a D, whether the flat area is large or small), which may prevent them from twisting relative to one another. For instance, a small flat area is provided or there is an oval shape at the interface, stress risers may be reduced in the plates, which may be particularly beneficial at higher rotations per minute. For example, in some implementations, rather than being bolted through or having a square or other shape with large protrusions, which may increase stress in the flywheel **152**, especially where the flywheel **152** is massive or spinning at a high rate, the clamping plate **320** to axle **608** interface may be shaped to induce very little stress into the axle **608** or clamping plate **320** while allowing torque to be transferred.

(131) FIG. **6**C illustrates a cross sectional view of an example multi-part flywheel **152**. As illustrated in the example implementation, a top clamping plate **320***a* may be connected with a top axle **608***a*. For instance, a top axle **608***a* may pass through the top clamping plate **320***a* so that the top clamping plate **320***a* may apply downward force on the axle **608**. In some implementations, the axle **608** may include multiple parts, such as an axle **608** portion and a washer (e.g., an axle ball washer **632**), where the washer (or a bottom portion of the top axle **608***a*) contacts a top-most stacking plate **322**. Accordingly, via the axle **608**, the top clamping plate **320***a* may apply force to the stacking plate(s) **322**. It should be noted that other configurations, such as direct contact or contact through another device are possible without departing from the scope of this disclosure. Accordingly, the clamping plate **320** may apply pressure at a center of the stacking plate(s) **322** via the washer and/or axle **608**.

(132) Similar to the description of the top axle **608***a* above, a bottom axle **608***b* may be coupled to a bottom clamping plate **320***b* and may apply force to a bottom-most stacking plate **322**. It should be noted that other configurations are possible, such as where the contact is direct, where the axles **608** are integrated with the clamping plates **320**, where the axles **608** are integrated with one or more stacking plates **322**, or otherwise.

(133) Additionally, force may be applied to a periphery (e.g., in an axial direction) or other area of the stacking plate(s) **322**. For example, bolts may be tightened down on the clamping plate(s) **320**, which apply force to an outer edge of the stacking plates **322**. The force may be applied via direct contact between the clamping plates **320** and the stacking plates **322** or via an intermediary device, such as a bushing or washer. In some instances, the clamping plates **320** may flex between the axle(s) **608** and the bolt(s) to provide the pressure. Accordingly, friction can be increased between the stacking plates **322**. In some implementations, the stacking plates **322** may be simple, solid plates rather than having perforations for fasteners in the plates, which may reduce strength and introduce stress risers due to centrifugal force, which may lead to increased complexity and failure modes. In other implementations, the stacking plates **322** may have perforations through which bolts may pass, which may increase a radius of the plates, provide simplicity in manufacturing, or increase an inter-plate (e.g., due to friction) force.

(134) As described below, the bolts may be tensioned to varying levels of tension to cause the friction force. Although different configurations are possible and contemplated, the flywheel **152** may include 8 bolts located around or proximate to a peripheral edge. Each bolt may be tightened to provide a defined torque or based on an applied force before the bolts are torqued (e.g., to apply a force of 2600 pounds per bolt), which may cumulatively provide a relatively even clamping and friction force across the stacking plates **322** (e.g., 16,000-21,000 pounds of clamping force).

(135) In addition to their roles in clamping together the clamping plates **320**, the bolts may include other features, such as the ability to mitigate failure of one or more stacking plates **322** (e.g., by catching a stacking plate **322** or portion thereof that slips or breaks). In some instances, the bolts may be replaced with other bolts of varying weights to assist in balancing the flywheel **152**. Other details and implementations are possible

and/or described elsewhere herein.
(136) It should also be noted that the top axle **608***a* and the bottom axle **608***b* should be aligned as perfectly as possible to reduce vibrations and improve alignment with bearings, etc. An assembly fixture and assembly procedure may be used to align the axles **608**. Although other implementations are possible, ball washers may be used with the axles **608** to allow some adjustability during assembly to improve alignment. It should be noted that flat washers or no washers (e.g., the axles **608** may be single components instead of broken into an axle body and axle washer) may be used.
(137) In some implementations, an axle-connection region of a clamping plate **320** may connect to an axle **608** in order to transfer force between the clamping plate **320** and the axle **608**. The connection may include a step that allows the clamping plate **320** to apply clamping force on the axle **608** (e.g., on a corresponding lip or step thereof), although the axle **608** may be integrated with the clamping plate **320** or the force may simply be applied onto the axle **608** by a bottom edge of the clamping plate **320** (e.g., where no step is included in the implementation). For instance, the axle-connection region of a clamping plate **320** may include a perforation in the clamping plate **320** through which the axle **608**, or a portion thereof, may pass. For example, an axle washer may be coupled with the axle **608** at the step, or a portion of the axle **608** itself may interact with the step.
(138) In some implementations, the axle **608** connection may include various shapes to the pass through that interact with corresponding shapes of the axle **608**. The perforation or a portion thereof may be oval shaped or have a flat or “D” shaped area, key, or other shape that allows torque to be transferred between the stacking plate **322** and the axle **608** (e.g., in addition to the torque that may be applied to the axle **608** by its contact with the top/bottom-most stacking plate **322**) without significantly increasing material stress at the connection point. In some implementations, this shape may be applied to the entirety of the perforation or only to a portion or step thereof (e.g., as in the illustrated step). This shape may be small, such as a ⅛.sup.th inch deviation in diameter or a flat section.
(139) Although other implementations are possible, the axle **608** and perforation diameter may be 3-5 inches. For example, a first (e.g., illustrated at a top of the figure) perforation/axle **608** diameter may be 3.75 inches. A second (e.g., illustrated downward from the first) perforation (e.g., step in the perforation)/axle **608** diameter may be 4.25 inches to allow force to be applied from the first diameter onto the axle **608** and then onto the stacking plates **322**. In implementations where the second step/perforation/axle **608** portion are oval shaped, the oval may vary from 4.375 inches to 4.250 inches, for example, although other implementations are possible and contemplated herein.
(140) The top axle **608***a* may couple with and/or extend through a top clamping plate **320***a* and a bottom axle **608***b* may couple with and/or extend through a bottom clamping plate **320***b*. As a clamping plate **320** may induce friction and transfer force to/from the stacking plates **322**. Accordingly, the clamping plates **320** may be designed to apply axial force to the plates without having high stress areas at the periphery where the clamping plates **320** may fail at high speeds. Accordingly, the clamping pressure may be increased and risk due to structural/material failure decreased. Example configurations of the clamping plates **320** are described elsewhere herein.
(141) Additionally, the clamping plate(s) **320** may include a connection area for coupling with the axle(s) **608**, which allows the rotational force to be transferred between the plates and the axles **608**. For instance, an axle **608** may extend through a clamping plate **320** and have or more shapes or structures that allow rotational, as well as clamping force, to be applied onto the staking plates (e.g., via the axle **608**). In some implementations, the axles **608** may have portions, washers, or ball washers that extend beyond an inner edge of the stacking plate **322** to apply force the stacking plates **322**. Although the washers are illustrated as being approximately the size of the passthrough in the clamping plates **320**, it should be noted that they may be omitted, combined with the axle body, be smaller radius than the passthrough, or be larger than the pass through (e.g., to apply force to the staking plates over a larger area).
(142) As the axles **608** or axle washers contact the stacking plates **322**, the application of clamping force by the bolts may cause the arms of the bend slightly and increase the force at the center that is applied by the axles **608**/axle washers. The thickness of the axle washer (or similar component) and the configuration of the clamping arms may be such that the distance between the ends of the arms (e.g., to the stacking plates **322**) may be minimalized when the plates are clamped. In some implementations, in addition or alternative to the clamping force at the center of the stacking plates **322**, the clamping plates **320** may apply clamping force along a peripheral edge of the stacking plates **322**.
(143) The clamping plates **320** may be constructed from aluminum, steel, or another material. For instance, the plates may be constructed from a ferromagnetic steel (e.g., AR500 steel plate) and may be stamped, formed, or machined into the desired shapes. Example masses of the clamping plates **320** may be 66-68 pounds when constructed from steel, although other implementations are possible.
(144) FIG. **6**D illustrates a side-top view of another example flywheel **152**, according to some implementations. In the example of FIG. **6**D, the shape of the clamping plates **320** has an X shape with two bolt holes proximate to the radial edge of each arm thereof. In the depicted example, the bolts may be angled as they pass from the top clamping plate **320***a*, through the stacking plates **322**, and to the bottom clamping plate **320***b*. By angling the bolts, rotational forces across the clamping plates **320**, stacking plates **322**, and axles **608** may be reinforced, which reduces the odds that the plates will move out of alignment when the flywheel **152** is spun up or down though the axle(s) **608**.
(145) In the depicted example, the bolts may be angled toward each other or away from each other on alternating clamping plate arms, which improves uniformity of force (e.g., circumferentially and axially) and rotational balance. For instance, in a first arm, the bolts are angled away from each other at the top plate, while, at a second arm 90 degrees from the first arm, the bolts are angled toward each other at the top plate, which pattern may repeat, as illustrated. Where the top clamping plate **320***a* and the bottom clamping plate **320***b* are the same, they may be rotated 90 degrees, so that the holes on each match the angles of the bolts. For example, a bolt may be perpendicular to a radial direction of the flywheel **152** and angled around the periphery, for example, at an angle to the axial direction of the flywheel **152**.
(146) In the depicted example of FIG. **6**D, the bolts extend through the top and bottom clamping plates **320***b* and through perforations in the stacking plates **322**. In the example implementation where the bolts are angled, the bolts may use wedge shaped washers that allow the force from the bolts to be applied to the clamping plates **320**. In some implementations whether with angled or straight (e.g., axial) bolts, the bolts and associated nuts may be tapered to allow them to extend partially into countersunk holes in the clamping plate(s) **320**.
(147) In the depicted example of FIG. **6**D, the top clamping plate **320***a* (and potentially the bottom clamping plate **320***b*) may be substantially flat on its top (partially as in **6**A or fully as in **6**D) and bottom surfaces, which allows the it to contact the stacking plates **322** and/or interact with a magnetic lifting component **352**. For example, a very flat top surface of the top clamping plate **320***a* that interacts with a magnetic lifting component may reduce eddy currents in the top clamping plates **320***a* caused by rotation relative to the magnetic lifting member **352**.
(148) FIG. **6**E illustrates a side-top view of another example flywheel **152**, according to some implementations. In the depicted example, the bolts extend axially through perforations **622** in the top clamping plate **320***a*, the stacking plates **322**, and the bottom clamping plate **320***b*. Depending on the implementation, the stacking plates **322** may have an equal quantity of perforations **622** as the quantity of bolts clamping the clamping plates **320**, the clamping plate(s) **320** may include additional perforations **622** proximate to their peripheral edge(s). These additional perforations **622** may be used in balancing the flywheel **152**, for instance, by drilling out the holes **622** or adding plugs to the holes. As noted elsewhere herein, there may be a space between one or both of the clamping plates **320** and the stacking plates **322**. For instance, the top clamping plate **320***a* and top stacking plate **322** may lack a space (e.g., as at **442**), which may prevent the top clamping plate **320***a* from flexing, thereby improving its flatness and interaction with a magnetic lift member. In some implementations, there may be a small gap/space between the bottom clamping plate **320***b* and a bottom-most stacking plate **322**, which allows some flex in clamping (e.g., to increase a force at the center/axles **608**). For instance, a portion of the bottom axle **608***b* and/or an axle washer (whether a ball washer or flat) may be used to provide a space between a center of the bottom clamping plate **320***b* and the bottom-most stacking plate **322**.

(149) In some implementations, the stacking plates **322** may include contours or scallops **610** around a peripheral edge, which may reduce failure points due to radial stress around bolt holes and/or assist with aligning the plates. For instance, a scallop **610** may be a scalloped shape or contour removed or omitted from a peripheral edge of a stacking plate **322**.

(150) The clamping plates **320** may be constructed from aluminum, steel, or another material. For instance, the plates **320** may be constructed from a ferromagnetic steel (e.g., AR500 steel plate) and may be stamped, formed, or machined into the desired shapes. Example masses of the clamping plates **320** may be 66-68 pounds when constructed from steel, although other implementations are possible.

(151) FIG. **6**F illustrates example flywheel axles **608***a* and **608***b*. Although other sizes and configurations are possible, FIG. **6**F illustrates a top and bottom axle **608***b* with axle washers **632** (e.g., ball washers). Depending on the implementation, the top and bottom axles **608***b* may be identical or have variations, such as their length, whether or not they include washers or axle washers **632**, whether they include a motor connection **666**. Some features of the axle **608** are described in reference to a single one of the top and bottom axle **608***b*, but they may be present on both or the other axle **608**.

(152) Depending on the implementation, an axle **608** may include a smooth shaft **664** (e.g., a 50-70 mm diameter shaft **664**) portion that interacts horizontally with one or more bearings to keep the flywheel **152** aligned. The shaft **664** may contact one or more seals to maintain the vacuum and may be polished to avoid friction with the seals.

(153) An axle **608** may include one or more bearing shelf (ves)/step(s) **668** that interact with bearings to provide vertical support to the flywheel **152** (e.g., to lift, lower, or hold it vertically).

(154) In some implementations, an axle **608** may include one or more clamping shelf (ves)/step(s) **670** that interact with a clamping plate **320**. For instance, the clamping step **670** could be a wider area than the shaft **664** so that the clamping plate **320** applies pressure on the clamping step **670** to hold the axle **608**. In some implementations, the axle **608** extends beyond the clamping step **670** and flywheel step **668**, so that the axle **608** applies pressure to a stacking plate **322**, as noted above. The contact with a stacking plate **322** may be via a washer, such as an axle washer **632** (which may be a flat or ball washer). The clamping step **670** may interact with an edge or corresponding step(s) on a clamping plate **320**.

(155) In some implementations, the clamping step **670**, an axle washer **632**, or another part of the axle **608** may be shaped to interact with a corresponding shape or structure in a clamping plate **320**. For instance, it may include a flat side, oval shape, protrusion, waves or ridges, teeth, or other structure that allows torque to be transferred between the axle **608** and the clamping plate **320** and/or stacking plates **322** (e.g., where a top or bottom stacking plate **322** includes a shape to match this structure). For example, as noted in further detail above, an oval or small flat side may be used to avoid stress risers in the material (e.g., of the clamping plate **320**).

(156) In some implementations, one or both of the axles **608** may include a motor connection **666** that may be a portion or extension of the shaft **664**. The motor connection **666** may include a flat, oval, D-shaped, or other structure/shape (e.g., a key or slot) that allows torque to be transferred between the axle **608** and another structure, such as a motor-generator **310** (e.g., via a magnetic coupling **318**, as described elsewhere herein). The motor connection **666** may additionally or alternatively include keys or other protrusions that improve the connection between the axle **608** and another structure (e.g., the magnetic coupling **318**, motor-generator **310**, etc.).

(157) Although a ball washer (at **632**) is illustrated on both the top and bottom axle **608***a* and **608***b* in FIG. **6**F, other implementations are possible and contemplated. For example, a ball washer may be used to provide a small amount of adjustability to the axle **608** alignment when top axle **608***a*, bottom axle **608***b*, stacking plates **322**, and clamping plates **320** are aligned. As illustrated in the example, two axles **608** may be used where the axles **608** are physically disconnected from each other. One or both of the axles **608***a* and **608***b* may lack a washer, may have a flat washer, and/or may have a ball washer.

(158) In some implementations, a ball washer may be flat on its bottom where it contacts a stacking plate **322** while it is rounded on a top where it contacts a corresponding curve in the axle body. Accordingly, the position of the axles **608** could be shifted slightly during assembly to allow the axles **608** to be positioned. As illustrated, in some implementations, a bolt may couple the axle washer **632** to the axle body in order to hold it in place during assembly.

(159) It should be noted that, in some implementations, flat washers or no washers are used with an axle **608**.

(160) FIG. **6**G illustrates cross section of an example bolt **614**, an example clamping plate **320**, several stacking flywheel plates **322**, and a bearing or bushing **612** that applies pressure from the clamping plate **320** on the stacking flywheel plate(s) **322** (e.g., at a peripheral edge, as noted above). In the depicted example, where the flywheel **152** uses stacking plates **322** and separate clamping plates **320**, various issues may be caused when the flywheel **152** spins at very high speeds. For example, at high RPMs (rotations per minute), the mass of the stacking plates **322** may cause them to stretch radially while thinning axially. Because some implementations of the clamping plates **320** have less mass at higher radiuses or a lower mass-to-cross section ratio, they may change shape less than the stacking plates **322** at high RPMs. Accordingly, to avoid structural problems, maintain friction among the stacking plates **322**, and reduce friction between components that change shaped differently, various mechanisms, such as Belleville™ washers **606**, bearings or bushings **612** between clamping and stacking plates, and/or other features. For instance, the bushings may be wear-resistant balls or cylinders.

(161) As noted above, in some implementations, the bolt shaft may contact the stacking plates **322** around the bolt **614**, may not contact the stacking plates **322**, or may contact at defined points (e.g., on a radially inward side, a radially outward side, etc.).

(162) As noted elsewhere herein, a bolt **614** may be positioned so that 2,000 to 3,000 pounds of force is on each bolt to create friction on the plates. As noted in reference to FIG. **6**G, a bolt **614** may contact the clamping plate(s) **320** via Belleville™ washers, dish springs, or other washers **606**, which may provide some spring in order to continue to apply pressure to the plates when they elongate or become thinner.

(163) For instance, a bolt **614** may not contact the stacking plates **322** unless there is a mechanical failure in the bolt **614** or the stacking plates **322**. Where a bolt **614** fails, the contour of the clamping location around the bolt **614** may catch the bolt **614** or fragments thereof. Similarly, where a stacking plate **322** fails, the bolt **614** may catch a stacking plate **322** or portions thereof to mitigate a more catastrophic failure or external damage.

(164) It should be noted that there may be a space between the clamping plate **320** and the top stacking plate **322** and the stacking plates **322** may each contact one another, which space may be larger during assembly, but the space may shrink as one or more of the clamping plates **320** are tensioned. In the depicted example of FIG. **6**G, the clamping plates **320** are in an unstressed position, so that the bearings or bushings **612** are shown unconnected from a top-most stacking plate **322**. It should be noted that this space is provided for visibility and that the clamping plate **320** may flex and/or be positioned differently, so that the bearing or bushing **612** contacts the stacking plate **322** and/or there is compression on the Belleville™ washers **606**.

(165) Depending on the implementation, an arm of the clamping plate **320** may directly or indirectly (e.g., via a bearing or bushing **612**) contact a top/bottom most stacking plate **322**. Because the bolts **614** may not allow the clamping plates **320** to move inward as the stacking plates **322** become thinner (although this effect is very small) at high RPMs, spring members, such as Belleville™ or cone washers **606** may be used to provide some flexibility and spring, which feature helps maintain the spacing and/or force between the clamping plate(s) **320** and the stacking plate(s) **322**.

(166) Accordingly, the Belleville™ washers **606** may keep a continual force on the plates even while shrinking while spinning. For example, the Belleville™ washers may provide a 1/100,000.sup.th or 1/10,000.sup.th flex that accommodates for thinning of the stacking plates **322**. Different spring members or washers may be used to increase the flex where thinning is greater at the clamping location, such as where a greater number of stacking plates **322** are used.

(167) Additionally, because the stacking plates **322** may stretch a greater amount than the clamping plates **320** at high RPMs, bearing or bushings **612** or other devices may be used to reduce radial friction between the clamping plates **320** and stacking plates **322** while maintaining clamping force and therefore friction between the stacking plates **322**. In some instances, the bearings or bushings **612** may be configured so that rotational friction between the stacking plates **322** and the clamping plates **320** is maintained. It should be noted that although "bushings" are referred to herein, other pivoting or movable structures may be used to allow the plates to expand relative to each other. For example, any device that allows contact while providing the ability to expand differently radially may be used. For example, a small amount (e.g., 5, 8, or 10 thousandths of an inch) of movement may be allowed by the bearing or bushing **612**. The movement may be due to rocking, rolling, pivoting, etc.
(168) For example, a bearing or bushing **612** may be a ball, cylinder, wheel, roller, or other piece that allows the stacking plates **322** to move relative to the clamping plates **320**. As illustrated in the example of FIG. **6**G, the bearing or bushing **612** may be a cylindrical piece of metal; although, it should be noted that other movable pieces that, such as those that roll or pivot back and forth are possible and contemplated herein. For example, a wear resistant metal ball or cylinder may be used.
(169) In some implementations, a bearing or bushing **612** may be partially recessed or otherwise held by a clamping plate **320**, such as in a recess, as illustrated in the cross section of FIG. **6**G. As example dimensions, the recess may be 0.3 inches in diameter and 0.2 inches deep, so that a ball may extend slightly; although the recess may be rectangular or another shape.
(170) Various quantities, configurations, sizes, or positions of bearings or bushings **612** and bushing holders may be used. For example, as illustrated in the example cross section of FIG. **6**G, a single cylindrical bearing or bushing **612** is shown radially inward from the bolt **614**. In other implementations, multiple bearings or bushings **612** may be used on each arm and, for instance, located to the sides (e.g., circumferentially) relative to the bolt **614**. For example, a single spherical bearing or bushing **612** may be located on each side of a bolt **614** at the clamping area of the clamping plate **320**.
(171) It should be noted that other implementations are possible and contemplated herein, such as where no bushings are used, where they are positioned or configured differently, or different mechanisms are used.
(172) FIG. **7**A illustrates a cross-sectional view of an example upper axle **608***a* disposed within an upper bearing assembly of a flywheel enclosure **304**. In the illustrated example, some components may be omitted for visibility. As illustrated, a top axle **608***a* may interact with a plurality of bearings **334** and/or seals (e.g., in a housing **722**) to provide vertical and/or horizontal support. For instance, multiple (e.g., **2**) bearings **334** may be used on an axle **608** to increase redundancy and safety. In some instances, a temperature sensor **704** or accelerometer may be located in or adjacent to the bearing housing **722**, which allows the flywheel **152** to detect a failure of one or more bearings **334** thereby increasing a safety margin. Other features, such as cooling loops (e.g., through which coolant may be circulated), vacuum connections, etc., may also be used.
(173) As illustrated, one or multiple seals may be located in or adjacent to the shaft of the axle. For instance, the seals may be housed within a bearing/O-ring housing **722** and contact the smooth sides of the axle shaft to seal a vacuum. In some instances, where the vacuum is actively established or maintained, the seals may change their shape by flexing inward to improve the seal. Similarly, the seals may be multiplied (e.g., doubled) for redundancy. Other structures, such as retaining clips may be located on one or both sides of the bearings **334**, so that they can be installed or replaced separately or with a housing **722**.
(174) In some implementations, the bearings and/or seals/shaft may be lubricated, for example, using a high durability and/or vacuum specific lubricant. In some implementations, a special material may be used for the seals to allow them to be used in a vacuum and/or without a separate lubricant. Depending on the implementation, the bearings may be dry bearings, such as a ceramic hybrid bearing, which beneficially reduces eddy currents and other issues due to moving in a magnetic field. Additionally, or alternatively, a dry film lubricant may be used for these components.
(175) In some implementations, as illustrated, an example magnetic lift member **352** may interact with (e.g., to attract) a flywheel **152**, such as a top clamping plate **320***a* (and/or stacking plates **322**). For instance, as illustrated, the magnet(s) of a magnetic lift member **352** may be located above, below, or next to the center of the axle **608***a*. For example, the magnets may be positioned by the magnetic lift member **352** (also referred to as the magnetic lift assist member/mechanism **352**) to closely interact with the flat area (e.g., **444**) of the top clamping plate **320***a*. For instance, the top bearing **334** may hold the top clamping plate **320***a*/flywheel **152** at a defined distance from the magnetic lift member **352**, so that a defined magnetic force is applied, which lifts the flywheel **152** wholly or partially. For instance, as described elsewhere herein, the magnetic lift may be less than (e.g., so that weight remains on a bottom bearing), equal to (e.g., so that weight is roughly balanced between the top and bottom bearings), or greater than (e.g., so that the top bearing is holding the flywheel **152** from being pulled closer to the magnet(s)) the weight of the flywheel **152** at the set distance.
(176) As illustrated and described in further detail below, the magnetic lift member **352** may be positioned close to the clamping plate **320***a*, which may be ferromagnetic (e.g., a magnetic steel) flat (or matching the shape of the magnetic lift member) shape. As shown, the flywheel **152** may be positioned at a center of the enclosure **304**, although other implementations are possible. As noted elsewhere herein, the magnets of the magnetic lift mechanism **352**/member may be stationary and coupled with the enclosure because magnets tend to be made out of weaker material that would not hold up well to rapid spinning (e.g., because rare-earth magnets, for instance, are mechanically weak). In the depicted example in FIG. **7**A, a cavity is shown in the magnetic lift member **352**, but this cavity may include one or more magnets, as described below. The magnetic lift member **352** may be assembled as a unit and then bolted or otherwise attached to a lid **328** of the flywheel assembly **150**.
(177) In the depicted example, an upper axle **608***a* may be coupled to a motor directly or via a magnetic coupling **318**, as described elsewhere herein.
(178) FIG. **7**B illustrates a cross-sectional view of an example lower axle **608***b* disposed within a lower bearing assembly **724** of a flywheel enclosure **304**. In the illustrated example, some components may be omitted for visibility. Similar to FIG. **5**A, as illustrated, a bottom axle **608***b* may interact with a plurality of bearings **466** and/or seals or structures to provide vertical and/or horizontal support. For instance, multiple (e.g., 2) bearing assemblies may be used on an axle **608***b* to increase redundancy and safety. In some instances, a temperature sensor or accelerometer may be located in or adjacent to the bearing housing **724**, which allows the flywheel **152** to detect a failure of a bearing thereby increasing a safety margin, improving efficiency, etc. Stacking plates **322** are also shown.
(179) As shown in the example of FIG. **7**B, a cap **442** is also shown. The cap **442** may seal (e.g., using gaskets and bolts) an interior cavity of the enclosure **304**. The cap **442** may provide access to move the flywheel **152** within the enclosure **304**; install, maintain, or adjust bearings **466** and seals; and perform other actions.
(180) In some implementations, the cap **442** and/or another component may be threaded, so that it may be twisted up/down, which adjusts the position of the bearings **466** and/or seals; or it may lift the flywheel **152** itself to set its position in the enclosure **304**. In other implementations, the flywheel **152** may be manually adjusted (e.g., to be at a defined distance from the magnetic lift member **352**) and then the bearings inserted or locked in position.
(181) FIG. **7**C illustrates an example flywheel **152** coupled with a portion of a magnetic lift member **352** and a lower bearing assembly outside of a flywheel enclosure **304**, for example, for purposes of illustration.
(182) As illustrated in the example of FIG. **7**C, an exterior of the magnetic lift member **352** has been omitted to show magnets **732**, which may be wedge magnets, and an example relative proximity to the top clamping plate **320***a*. For instance, the wedge magnets **732** may, when in an active configuration, pull on the flat area (e.g., **444**) of the clamping plate **320***a*, although other implementations are possible and contemplated. It should be noted that although the magnets **732** and other components of the magnetic lift member **352** are illustrated floating above the top clamping plate **320***a* (e.g., instead of attached to an enclosure **304**/lid **328**) for purposes of illustration.

(183) The example of FIG. **7**C also illustrates a lower bearing assembly **724** that holds one or more bearings at the bottom of the flywheel enclosure **304**. For instance, a lower bearing assembly **724** may be welded, integrated with, or bolted to an enclosure tub **326**. The lower bearing **724** may support none, a portion, or all of the weight of the flywheel **152**. In some implementations, the lower bearing **724** may merely be present to keep the flywheel **152** horizontally aligned.
(184) The lower bearing **724** may include a shipping support area **462**, such as a shipping ring, on which the weight of the flywheel **152** may rest during shipping, storage, or when not in use. The shipping support area **462** may be any device that may support the flywheel **152**, such as a plastic or metal ring in the enclosure tub **326**.
(185) The height and/or relative positioning of the bearings may also be adjusted because a quantity (e.g., 10, 14, 18, 28, or other quantities) of stacking plates **322** may vary, and thicknesses of each plate may vary (e.g., by a thousandth of an inch), the overall thickness of the flywheel **152** may vary enough to affect the functioning or longevity of the bearings unless there is flexibility in the design, as illustrated, to accommodate different heights.
(186) As described elsewhere herein, a retaining cap **442** or another mechanism may seal the enclosure and/or capture an adjustment nut so that it does not accidentally move in order to lock the Z/vertical axis of the flywheel **152**.
(187) FIGS. **8**A-**8**D illustrate various views, components, and constructions of an example flywheel positioning system. The flywheel positioning system may be configured to position the flywheel **152** within the enclosure **304**, as described elsewhere herein. Although a certain implementation is described, other implementations and features are contemplated, and the provided examples should be understood as examples.
(188) In some implementations, the flywheel positioning system may adjust the position of the flywheel **152** between the top and bottom bearings **466** so that a distance between the bearings **466** may be adjusted to match a size of the flywheel **152** and thereby to minimize wear on the bearings **466** while using their functionality. For instance, as noted elsewhere herein, the flywheel positioning system may lift the bottom bearing **466** upward, in turn lifting the flywheel **152** upward, until the flywheel **152** contacts the top bearing and/or is correctly distanced from the magnetic lift member/mechanism. In some implementations, as the flywheel **152** is lifted up, it may contact a surface, such as a top bearing (e.g., at the top of an enclosure **304**, magnetic lift member **352**, or bumper. Once it contacts the surface, it may be backed down by a defined amount to correctly position the flywheel **152**. In some implementations, the positioning system (e.g., a nut bearing holder **464**) may include one or more marks that may be used to determine correct positioning.
(189) In some implementations, the flywheel positioning system may be used to move the flywheel **152** or components of the flywheel assembly **150** between modes. As noted above, the flywheel positioning system may move the flywheel **152** (e.g., via a bottom bearing **466**) or a shipping surface/ring **462** in order to move the flywheel **152** between a shipping position and an active position.
(190) For example, as noted in further detail elsewhere herein, a flywheel **152** may rest on a shipping ring **462** during shipping or storage. The flywheel positioning system may be adjusted to lift the flywheel **152** off the shipping ring **462** and into an active position, for instance, by engaging a bottom bearing **466** (e.g., by moving the bearing **466** and/or flywheel **152**) and/or the top bearing (e.g., by moving the bearing and/or flywheel **152**).
(191) FIG. **8**A illustrates a cross sectional view of an example flywheel positioning system, which may include a retaining cap **804** that couples with a positioning nut/nut bearing holder **464** and a flywheel enclosure **304**. For example, the base ring **440** may be integrated or coupled with a bottom plate **436** of the enclosure **304** tub. One or more bottom ribs **438** may also be coupled with the bottom plate **436** and base ring **440** (e.g., by a weld), which provides strength to the enclosure **304** sufficient to hold the weight of the flywheel **152**.
(192) In some implementations, the base ring **440** may be threaded on its interior or exterior to interact with a nut bearing holder **464**, as described below. In some implementations, a retaining cap **804** may be coupled with the base ring **440** via one or more bolts, which bolts may provide adjustability to the rotation of the retaining cap **804** on the enclosure **304**, as noted below.
(193) In the depicted examples, a flywheel **152** and flywheel axle **608***b* are shown inside an enclosure **304**. Although the flywheel positioning system could be used with a top axle **608***b*/the top of a flywheel, the illustrated examples of FIG. **7**B illustrates the flywheel positioning mechanism used with a bottom axle **608***b* of the flywheel **152**.
(194) As shown, an axle **608***b* may interact with one or multiple bearings **466** held by a lower bearing holder **464**. The lower bearing holder **464** may be a nut bearing holder **464** where the nut includes a bearing **466** holding portion, a tightening portion, and one or more threads, as described below. The bearing(s) **466** may support the axle **608***b* of the flywheel. In some instances, the bearing(s) **466** and/or nut bearing holder **464** may be held by a lower sleeve **808**.
(195) As illustrated in the example, the nut bearing holder **464** may hold the bearings **466** and may rotate (e.g., using threads) within a lower ring of the enclosure **304** tub to move the nut bearing holder **464** and bearing(s) **466** upward or downward relative to the enclosure **304**, which may, in turn, move the flywheel upward or downward. For example, when in a shipping configuration, the nut bearing holder **464**, bearing(s) **466**, and flywheel **152** may be moved downward so that the bottom surface of the clamping plate **320***b* rests on the shipping ring **462**, which may be a metal, plastic, or another material on which the flywheel **152** may rest to remove stress from the bearing(s) **466**. In some instances, the flywheel **152** may rest directly on the bottom of the enclosure **304** tub when in a shipping or storage position.
(196) Although other implementations are possible, a nut locking mechanism may include one or more of a cap **804**, a hex lock **806**, and/or a nut bearing holder **464**, etc. In some implementations, as described in further detail below, a cap hex lock **806** may include various protrusions, recesses, or other structures, such as teeth **814** (e.g., defining a set of angles at which a nut may be held), that interact with the nut bearing holder **464** to prevent the nut bearing holder **464** from twisting relative to the enclosure **304**, which may change the vertical position of the flywheel in the enclosure **304**. The cap **804** hex lock **806** is described in further detail below.
(197) In the depicted implementation, a retaining cap **804** is also shown. The retaining cap may be an implementation of a cap **442**. The retaining cap **804** may include one or more O-rings or channels to seal the vacuum internal to the enclosure **304**. Accordingly, when the retaining cap **804** is placed onto the enclosure **304** (e.g., after flywheel positioning), a vacuum may be maintained.
(198) In some implementations, the retaining cap **804** may hold the cap hex lock **806** in position on the nut bearing holder **464**, so that the nut bearing holder **464** cannot rotate when the retaining cap **804** is bolted to the enclosure **304** (e.g., the base ring **440**). For example, when attaching the retaining cap **804**, it may be rotated to mate up with the cap hex lock **806** to hold it in a specific position, causing the cap hex lock **806** to bridge the space between the nut bearing holder **464** and the retaining cap **804** and prevent the nut from rotating. It should be noted that although the cap hex lock **806** is described as a separate device from the retaining cap **804**, it may be integrated with either the nut bearing holder **464** or the retaining cap **804** to simplify installation. Example implementations of the nut bearing holder **464**, cap hex lock **806**, and retaining cap **804** are described elsewhere in further detail below.
(199) Accordingly, as an example procedure for changing the mode of the flywheel from a shipping position to an active position, a technician may rotate the nut bearing holder **464**, so that it moves upward and lifts the flywheel internal to the enclosure **304** and off of the shipping ring **462**. Once the flywheel **152** (e.g., a top clamping plate **320***a*) contacts a surface at the top of the enclosure **304**, magnetic lift member **352**, top bearing(s) **466**, bumper, or other component, the technician may stop rotating the cap **804** hex lock **806** upward and may back it off slightly to relieve pressure on the top bearing **466** or contact with another component. The technician may then insert a cap hex lock **806** around the hex head of the nut bearing holder **464** to mesh the teeth **814** of the lock with the nut. The technician may then place the retaining cap **804** onto the base ring **440** and rotate it until it interacts with the cap hex lock **806** (as described below) to hold the cap hex lock **806** in a given orientation so that it does not rotate. The technician may then tighten bolts between the retaining cap **804** and the base ring **440** to seal the enclosure **304** and lock the nut

bearing holder **464** in place.
(200) Accordingly, the position of the flywheel **152** and lower bearing **466** may be adjusted to accommodate for shipping, variations in flywheel thickness, or other aspects. By locking in an adjustable vertical position of the flywheel **152**, the flywheel positioning system prevents unintended movement of the nut bearing holder **464** during flywheel rotation or vibration. It also allows the position to be adjusted in the future for troubleshooting or maintenance, where adhesives or welds would not allow such access. Accordingly, the lock is secure, sealed, and accessible.
(201) Although not visible in FIG. **8**A, one or more sensors may also be located in the lower bearing assembly, such as in the base ring, adjacent to the bearings **466**, in the retaining cap **804**, or otherwise. The sensors may measure vibration, temperature, rotational velocity, or otherwise.
(202) FIG. **8**B illustrates a bottom-up view of an example nut bearing holder **464** coupled with a flywheel enclosure **304** and a hex lock **806**. As illustrated in the example of FIG. **8**B, a nut bearing holder **464** is located in a base ring **440** and a retaining cap **804** is removed for visibility. As illustrated, the nut bearing holder **464** may be in a defined rotation that affects the height of the flywheel **152**. Because small changes in the height of the flywheel **152** may affect its functioning (e.g., while it interacts with a magnetic lift member **352** or bearing(s) **466**), a few degrees of rotation of the nut bearing holder **464** can significantly affect the longevity, etc., of the bearings **466**. Accordingly, the hex lock **806** may include teeth **814** on a radially inward edge, any of which may interact with the sides of the nut bearing holder **464** in various rotational positions. As noted below, the hex lock **806** may interact with a retaining cap **804** to hold the hex lock **806** in place, so that the hex lock **806** bridges the gap and/or rotational difference between the nut bearing holder **464** and the retaining cap **804**. Accordingly, when the retaining cap **804** is placed on it, the cap **804** and hex lock **806** combination captures the torque of the nut bearing holder **464**, which may be otherwise disposed to rotate due to the rotation of the flywheels **152** and bearings **466**.
(203) Also, as illustrated in FIG. **8**B, the nut bearing holder **464** may include a hex shape to interact with wrench or other tools of various sizes, although it may be a different shape to accommodate different tools. In some implementations, it may include one or more slots at the bottom, so that a flat tool or a rod can be used to tighten or loosen the nut.
(204) FIG. **8**B illustrates an example cap **804** or nut hex lock **806** (also referred to as a hex lock **806**). As illustrated in the example of FIG. **8**B, the hex lock **806** may be a ring that may extend around a hex nut, such as the nut bearing holder **464**; although, it should be noted that it may extend only partially around the nut. The hex lock **806** may be a plate or disk of material that is formed, machined, or stamped (e.g., from steel) to have its features, as noted below.
(205) In the depicted implementation, the hex lock **806** may include teeth **814** or other grooves, protrusions, recesses or other structures disposed around an inner edge of the ring. For instance, the angles of the teeth **814** may match the angle of the hex lock **806** (e.g., 120 degrees) and where the radius of the nut bearing holder **464** matches the radius of the corners of the teeth **814**, so that when the hex lock **806** is placed on the nut, the teeth **814** hold the nut.
(206) The quantity of teeth **814** of the hex lock **806** may be varied depending on the increments of angles at which the nut may be held and adjustability of the retaining cap **804**. For instance, the teeth **814** may be larger and therefore fewer if the precise angle of adjustment of the nut is less important but smaller and greater in number of more precision is required for the application. Similarly, as noted elsewhere herein, the retention cap **804** may provide adjustability in its mounting to the enclosure **304** and/or interaction with the hex lock **806**, so the teeth **814** may be made larger based on the retaining cap **804** adjustability.
(207) The hex lock **806** may also include one or more protrusions **812** at an outer edge of the ring (e.g., a full or partial ring) that may interact with another structure to prevent the hex lock **806** from rotating. For instance, the hex lock **806** protrusions **812** may interact with corresponding protrusions or recesses of the retaining cap **804**, as noted below, to hold the hex lock **806** in a position. The hex lock **806** may include various quantities of protrusions, such as the six pairs of protrusions illustrated in the example. The protrusions and their interaction with one or multiple positions and structures are described in further detail below.
(208) It should be noted that, although nut interaction teeth **814** are illustrated in the inner edge of the ring and hex lock protrusions **812** are illustrated on the outside of the ring, other implementations are possible and contemplated herein. For instance, the hex lock **806** and/or retainer cap **804** may interact with an inner surface of the nut bearing holder **464** to keep it in place.
(209) As noted below, in some implementations, the structure of the hex lock **806**, such as the teeth **814**, may be integrated with the retaining cap **804**.
(210) FIG. **8**C illustrates a top view of an example retaining cap **804**. As illustrated, the retaining cap **804** may include hex lock interaction elements **828**. For instance, the interaction elements **828** may be raised above the body of the retaining cap **804** in order to extend upward into the space next to a nut bearing holder **464** and interact with the hex lock **806**. For instance, the retaining cap **804** may include an equal or fewer number of hex lock interaction elements **828** as the hex lock **806** includes locking protrusions **812**, so that these components interact with each other in order to rotationally couple the hex lock **806** to the retaining cap **804**. As illustrated in the example, the hex lock **806** may include more locking protrusion **812** pairs than the retaining cap **804** includes interaction elements (e.g., a 2:1 ratio), so that the hex lock **806** can be locked into multiple positions relative to the retaining cap **804**. In some instances, these positions may be offset at various angles to provide further adjustability to the interaction.
(211) In some implementations, the retaining cap **804** may include raised areas **826** that contact the bottom of the hex lock **806** to hold it in place. For instance, the top of the hex lock **806** may press against another area of the nut bearing holder **464**, base ring **440**, or another area to keep it from falling off the hex lock interaction elements **828**.
(212) In some implementations, the retaining cap **804** and/or base ring **440** may include recesses or other areas that may hold seals (e.g., gaskets or O-rings) in order to seal the vacuum internal to the enclosure **304**. For example, two grooves are illustrated as O-ring holders in FIG. **8**C. In some implementations, the retaining cap **804** may include other components or structures, such as a vacuum hose adapter to which a vacuum hose may be attached to pull the vacuum in the enclosure **304**.
(213) In some implementations, the retaining cap **804** may include adjustable bolt holes or slots through which the retaining cap **804** may be bolted to the enclosure **304**, such as the base ring **440** as noted above. For example, the retaining cap **804** may be bolted at various positions, which increases adjustability of the retaining cap **804**, hex lock **806**, and nut bearing holder **464** combination, so that more precise rotations and, therefore, heights may be locked.
(214) As illustrated in the example, a nut bearing holder **464** may be rotated to move it upwards or downwards in the base ring **440** and/or lower sleeve **808**. The nut bearing holder **464** may hold a bearing **466**, which, in turn, holds the lower/bottom axle **608***b* of the flywheel **152**. In some implementations, the nut bearing holder **464** may be open at a center/bottom so that the bottom axle **608***b* is visible, although other implementations are possible and contemplated herein. Once the nut bearing holder **464** is rotated to a correct rotation/height, the hex lock **806** and/or retaining cap **804** may be placed thereon to secure its rotation and therefore the vertical/Z height of the flywheel **152**.
(215) FIG. **8**D illustrates a top view of an example nut bearing holder **464**. The example nut bearing holder **464** may have the illustrated configuration, although other configurations are possible. Additionally, the nut bearing holder **464** is illustrated without threads for clarity but, in some implementations, it may include threads on an outer face to allow it to be twisted up or down. The bearing(s) **466** are also removed from the bearing holder in order to illustrate an example structure of the nut bearing holder **464**.
(216) The nut bearing holder **464** may have a hex interface and/or notches to allow it to interact with tools and/or the hex lock **806**. The hex interface may be located at the bottom of the nut bearing holder **464**, for example, below the outer face and bearing holder **464**.
(217) The nut bearing holder **464** may have a bearing holder body with an outer face and a bearing holder portion on the inside that holds one or

more bearings **466**. For instance, the bearing holder body may be cylindrical with threads on a radially outward face and a hollowed-out core configured to hold one or more bearings **466**. The core or center of the bearing holder portion may support vertical and/or horizontal bearings **466**, such as on one or more bearing steps. In some implementations, the bearing holder body may include a cavity at a center, such as is illustrated, to hold the flywheel axle **608***b* or let it pass therethrough.

(218) The nut bearing holder **464** may be resized, elongated, or widened based on the size of axle **608***b* and/or tightening tool being used. Furthermore, other configurations are possible and contemplated, such as where the nut bearing holder **464** additionally or alternatively holds a shipping ring **462**, where the nut bearing holder **464** has a different shape, or otherwise.

(219) The bearing(s) **466** may move upward or downward in an aperture of an enclosure **304** based on rotation of the nut bearing holder **464**.

(220) As shown, the bearings **466** may be held at the center of an aperture/hole in the enclosure **304** bottom. The enclosure **304** bottom may include a base ring **440** coupled with a bottom plate **436** and bottom rib **438** to provide strength to support the flywheel **152**. In some implementations, the base ring **440** may include a lower sleeve **808** that is positioned between a portion of the nut bearing holder **464** and the base ring **440** to provide threads and/or reduce vibrations (e.g., from a space around the nut bearing holder **464**). In some implementations, the base ring **440** or other components may be coupled with a bolt plate at the bottom of the enclosure **304** tub, which may securely hold the lower bearing **466** assembly in place.

(221) In some implementations, the nut bearing holder **464**, lower sleeve **808**, space around the nut bearing holder **464**, or other area of the lower bearing **466** assembly may include one or more sensors that relay bearing **466** or flywheel **152** health/status to the flywheel CPU/CNS/controller. For instance, a temperature sensor and an acceleration sensor may be placed at the bearing **466** to detect bearing **466** wear and/or failure or other anomalous conditions. Wiring for the sensors may pass through a retaining cap **804** or through the side of the enclosure **304**, as illustrated elsewhere herein.

(222) FIG. **8**D illustrates a top view of an example cap hex lock **806**, and nut bearing holder **464** in an assembled position, but with the flywheel **152** and flywheel enclosure **304** omitted for clarity.

(223) The example raised area(s) **826** of the retaining cap **804** are illustrated supporting the cap hex lock **806** to keep it in place. Hex lock interaction element(s) **828** of the retaining cap **804** are also shown interacting with locking protrusions **812** of the hex lock **806**. For instance, the interaction elements **828** are held in between pairs of the locking protrusions **812** around an outer edge of the cap hex lock **806**. It should be noted that different configurations, such as pairs of protrusions being on the cap **804** instead of or in addition to the hex lock **806** are possible. Similarly, other structures in which the components are rotationally locked are possible and contemplated herein.

(224) It should be noted that although the cap hex lock **806** and retaining cap **804** are illustrated as separate components, in some implementations, they may be integrated into a single component. For instance, an inner diameter of the retaining cap **804** may include teeth **814** configured as the hex lock **806**. Accordingly, while separating these components may provide easier manufacturability and assembly, they may be combined into a single unit designed to hold the nut bearing holder **464** in place, for example, at various rotations and corresponding heights.

(225) FIG. **8**D illustrates a top view of an example retaining cap **804**, cap hex lock **806**, and nut bearing holder **464** in an assembled position, but with the flywheel **152** and flywheel enclosure **304** omitted for clarity. In the depicted example, the assembled retaining cap **804** and cap hex lock **806** are shown in a locking position to prevent the nut bearing holder **464** from rotating.

(226) In addition to the interactions described above between the retaining cap **804** and the cap hex lock **806**, the teeth **814** of the hex lock **806** are shown interacting with the corners of the hexagonal/hex interface of the nut bearing holder **464** to rotationally lock the hex lock **806** with the nut bearing holder **464**.

(227) The retaining cap **804** may be bolted with the base ring **440** or other component of the flywheel enclosure **304**. For example, a technician may position the nut bearing holder **464** (e.g., as a correct height, as noted above), place the cap hex lock **806** in the retaining cap **804**, and then place the retaining cap **804** onto the base ring **440** while rotating it slightly (e.g., using adjustability provided by the bolt holes/slots) until the teeth **814** of the hex lock **806** fall into place with the nut bearing holder **464**. The bolts of the retaining cap **804** may then be tightened to the base ring **440** of the enclosure **304** to secure the entire assembly in the set position and, in some instances, seal an internal vacuum.

(228) FIG. **9**A illustrates an example assembly for an upper bearing of a flywheel assembly **150** including a magnetic motor coupling **318**, top axle **608***a*, upper bearing seal/O-ring housing **722**, and magnetic lift member **352**. It should be noted that although these components are illustrated as being separate, they may be combined or further separated. For instance, a bearing housing **722** may be combined with the magnetic lift member **352** and/or the magnetic coupling **318**.

(229) As illustrated in the example of FIG. **9**A, the top axle may pass through a magnetic lift member **352**, and which may be a ring coupled to the lid of the flywheel enclosure **304** (e.g., to an underside thereof). For instance, the magnetic lift member **352** may be positioned at the bottom-most point of the upper bearing assembly in order to be as close as possible to the flywheel **152** (e.g., to the top clamping plate **320***a*) to maximize the magnetic pull. For instance, where a flywheel **152** weighs 1800 pounds, at a defined distance, the magnetic lift member **352** may pull with a force of 1500 pounds, so that a bottom bearing only holds 300 pounds. In another configuration, where a flywheel weighs 1800 pounds, the magnetic lift member may pull with a force of 2100 pounds, so that the top bearing holds 300 pounds pushing upwards.

(230) The upper bearing O-ring housing **722** may hold one or more seals and/or one or more bearings **466**. For instance, the bearings **466** and seals may interact with the top axle **608***a*. The bearings are described elsewhere herein and may prevent the top axle **608***a* from moving upwards (e.g., due to the magnetic lift member **352**) and/or may keep the axle **608***a* from wobbling around an axis. The bearings **466** may be doubled for redundancy, as noted above. In some instances, the upper bearing O-ring housing **722** may also include one or more sensors that detect issues with the bearing(s).

(231) The upper bearings **466** may be oriented to reduce friction and/or provide support horizontally and/or vertically. Although the bearings **466** may be doubled to tripled for redundancy, minimal pressure should be applied to the bearings by balancing the flywheel **152** and/or using support mechanisms, such as the magnetic lift member **352**. The bearing(s) **466** may be ceramic/ceramic hybrid bearings and/or could use dry film lubricant.

(232) The upper bearing O-ring housing **722** may additionally or alternatively house one or multiple seals that interact with a shaft of the top axle **608***a*, and/or seals may be incorporated with bearing(s) **466**, as noted above. In some instances, the seals may be integrated with another component, such as a motor-generator housing, magnetic lift member **352**, enclosure lid **328**, and/or otherwise.

(233) The upper bearing O-ring housing **722** may be coupled to a top of the enclosure lid **328**, to the top of the magnetic lift member **352**, to the bottom of the magnetic coupling **318**, or to another component of the flywheel assembly **150**.

(234) The magnetic coupling **318** may be located near the motor-generator **310**, for example, near an end of the top axle **608***a* in order to interact with the top axle **608***a*, for instance, a lid **328** may be located between the housing **722** and the magnetic lift member **352**. As described elsewhere herein, the magnetic coupling **318** may interact with the shape, groove, machine key **916**, or other components of the top axle **608***a* to transfer torque to/from it.

(235) Example implementations of the magnetic coupling **318** are described below and may include various components, such as an external rotor top **912** and bottom **410**, as well as an internal rotor top **914** and bottom (not visible) that may rotate within the external rotor top **912** and/or bottom **410**. For instance, the external rotor top **912** and/or bottom **410** may be coupled (e.g., bolted) to a motor mount, upper bearing O-ring housing **722**, enclosure lid **328**, or otherwise, as described elsewhere herein.

(236) As illustrated in FIG. **9**A, the top axle **608***a* may include a machine key **916** that improves the connection with the magnetic coupling **318**.

(237) As illustrated in the example, the bearings **466** may be held at the center of the magnetic lift member **352**, for example, by an upper bearing holder **904**. The upper bearing holder **904** may be held by the magnetic lift member **352**, the upper-bearing O-ring housing **722**, an enclosure lid **328**, etc. For example, the upper bearing holder **904** may be located within or above a ring of the magnetic lift member **352**. The magnetic lift member **352** may be bolted to the bottom of the enclosure lid **328** (e.g., using the illustrated bolts).
(238) Depending on the implementation, the magnetic lift member **352** may include a mag lev backer **906** (also referred to as a backer ring) to which magnets may be attached or otherwise supported, as described below. In some implementations, the magnetic lift member **352** may additionally or alternatively include a magnet holder **908** that may couple with the mag lev backer **906** and/or magnets. For instance, the magnet holder **908** may be a cap or structure that overlaps over the bottom of the magnets to help support them and/or is bolted to a mag lev backer **906**.
(239) As illustrated, a magnet holder **908** may extend under the magnets to protect them and/or hold them to the mag lev backer **906** or another component. In the illustrated example, an upper bearing holder **904** may be coupled (e.g., bolted or welded) to the magnetic lift member **352** and may provide horizontal and/or vertical support to the bearing(s) **466**. The mag lev backer **906** may be a flat plate or disk with bolt holes to which the magnet holder **908** may attach. Similarly, the mag level backer **906** may bolt to a lid **328** of an enclosure **304** (not shown in FIG. **9**A). The magnet holder **908** may support the magnets or merely cover them, although other implementations are possible. The magnet holder **908** and mag lev backer **906** may be circular, have protrusions (e.g., for bolts), or have other shapes. Similarly, these components may be bolted at a center, periphery or otherwise supported.
(240) FIG. **9**B illustrates a bottom view of example internal components of a magnetic lift member **352**. As illustrated, the magnet holder **908** and upper bearing holder **904** have been omitted to show an example internal structure of the magnetic lift member **352**. In the illustrated example, a magnetic lev backer **906** is shown coupled with a plurality of magnets **732**. The magnets **732** may be glued to the mag lev backer **906** and/or held by a magnet holder **908**, for example.
(241) The mag lev backer **906** may be a strong ring (e.g., a steel ring), that holds the magnets **732**. The ring may include bolt or other connection points that couple with an upper bearing holder **904**, a magnet holder **908**, and/or an enclosure **304** (e.g., the enclosure lid **328**) as described and illustrated in the examples herein.
(242) Bolts are illustrated extending through various components herein, such as in the mag lev backer **906**. In some implementations, the bolts may be the same length as the bolt holes in which they are disposed, for example, to reduce gaps or other non-continuous structures, which may further cause inconsistencies with eddy currents, magnetic flux, or otherwise. For instance, the bolts may be trimmed to be the same size and length as the holes.
(243) The magnets **732** may be wedge magnets disposed around the mag lev backer **906** with a flux pointing downward to maximize interaction with the flywheel **152**. For example, as illustrated, the magnets **732** may be on a stationary component of the flywheel **152** instead of a rotating component because magnets may be constructed from weaker materials that may break under the stress of high rotational velocities. Additionally, by placing the magnets **732** on a stationary component, eddy currents and resulting forces may be reduced.
(244) In some implementations, a top clamping plate **320***a* may be constructed from laminated cores of insulated sheets of metal, or made from a non-conductive material, to reduce eddy currents in the top clamping plate **320***a*, although other implementations are possible, as described elsewhere herein.
(245) FIG. **9**C illustrates a top-down view of another example magnetic lift member **352**. In the depicted implementation, backer plate **906** and magnet holder **908** are shown. A bearing **466** held by an upper bearing holder **904** around an upper axle **608***a* is also shown. A retaining clip **922** is shown around the upper axle **608***a* is shown holding the bearing(s) **466** in place.
(246) In the depicted example, a cooling assembly **924** is also shown around or integrated with the bearing holder **904**. For instance, the cooling assembly **924** may have a chiller ring **926** that allows coolant to circulate around a bearing holder **904** or another component. The chiller ring **926** may be coupled with a coolant tube **928** that receives coolant, delivers it to the chiller ring **926**, and removes it to a pump and/or radiator (not visible) external to these structures.
(247) In the illustrated example, a connector assembly **932** or connector seal plate is also shown with barbed tube connectors and an electrical connection. The connector assembly **932** may receive pneumatic tubes for a vacuum assembly **308**, pneumatic tubes for a chiller ring **926**, an electrical connection for sensors, and/or other components. While the connector assembly **932** is illustrated floating above the magnetic lift member **308**, it may be coupled or integrated with a lid **328** or various other components. Other components, such as support brackets, seals, motors, etc., are omitted from FIG. **9**C for purposes of illustration.
(248) FIG. **9**D illustrates example components and constructions of an example flywheel-motor coupling, such as a magnetic coupling **318** that couples the motor-generator **310** to the flywheel **152**. Although a magnetic coupling **318** is described, it should be noted that a physical, direct coupling, a clutch, a gear or gearbox, etc., may be used. As illustrated, the magnetic coupling **318** may be positioned at the end of a top (or bottom, in other implementations) axle **608** to interact with the flywheel **152**.
(249) The magnetic coupling **318** may allow mechanical interaction between the motor-generator **310** and the flywheel **152**. For example, as described below, an inner rotor **964** (which may include the inner rotor **914** noted above) and outer rotor **962** (which may include one or more of the top outer rotor **912** and bottom outer rotor **410** noted above) may each include magnets **952** that interact with each other, to transfer force. The magnetic coupling **318** may provide a dampening effect between the motor **310** and flywheel **152** to avoid transfer of vibration or of jerk. For example, alternating magnets **952** may be used, which serve as gears using the shear effect of the magnets **952** and may have a dampening effect.
(250) As noted above, an example magnetic coupling **318** may be coupled with a top axle **608***a*, which passes through a magnetic lift member **352** and upper bearing housing **722**, depending on the implementation. As illustrated, the magnetic coupling **318** may be located at the end of the top axle **608**. The upper bearing O-ring housing **722** is also omitted from FIG. **9**D to show other components.
(251) In some implementations, a magnetic coupling **318** may be assembled in a single unit that may be bolted to a motor mount (e.g., **406**) and/or to a flywheel enclosure **304**. Accordingly, it can be pre-assembled, and the manufacturing process can be improved for both safety and speed.
(252) FIG. **9**D illustrates an example magnetic coupling **318** with an exterior rotor top (e.g., **912** and top component **914**) removed to expose the internal structure of the magnetic coupling **318**. As illustrated in the example implementation, a series of magnets **952** may be disposed at an internal rotor **964** and an external rotor **962**. Cylindrical magnets (e.g., N52 neodymium magnets) may be used and placed in the rotors **964** and/or **962**, so that their poles alternate, although other implementations are possible. In some implementations, the magnets **952** may be rectangular or another shape or configuration. Similarly, although the rotors **964** and **962** are radially located from each other, they may be vertically (e.g., where the magnets of opposing rotors are located vertically instead of radially) or otherwise oriented.
(253) The magnets **952** may create a shearing force against each other that serves to couple the rotors **962** and **964** together. For example, opposite magnets could be north and south oriented, so that they pull towards each other. In some implementations, where the magnets **952** are offset from each other, a shear force may be maximized to couple the two rotors **962** and **964** together.
(254) In some implementations, a seal or membrane may be placed between the two rotors **962** and **964** to maintain the vacuum internal to the enclosure **304**.
(255) As illustrated, the magnetic coupling **318** may include a number of bolts coupling various portions together. For instance, one or more bolts may couple the internal rotor **964** bottom and/or internal rotor shunt **972** (e.g., coupled or integrated with the internal rotor **964**) to another component, such as a motor-generator **310** or a top axle **608***a*. Similarly, other bolts may couple together external rotor shunt(s) **974** (e.g., coupled or integrated with the internal rotor **962**), and other components. The bolts may be used to adjust the positioning of the magnetic coupling **318**.

(256) As illustrated, the magnetic coupling **318** may include an internal shunt **972** and an external shunt **974** or backer rings (e.g., a steel backer ring) that provide structure and support to the magnets **952**, and/or they may improve magnetic flux. For instance, the shunts or rings may cause the magnetic flux to be more controlled and focused. In some instances, these components also increase the strength of the assembly.
(257) In some instances, the shunts **972**/**974** or other devices or lips that hold the magnets **952** in the rotors **962**/**964** may be reinforced as the rotors may not only experience hundreds or thousands (e.g., 2500 pounds) of force due to the magnetic pull but may also experience significant centrifugal force while the flywheel **152** is spinning (e.g., at 12,500 RPM). In order to reduce radial forces on the magnets **952**, they may be placed near to the axis of rotation, as illustrated.
(258) Although not illustrated, in some implementations a motor-generator may be coupled with the internal rotor top (e.g., at **914**). In other implementations, the motor-generator may be directly coupled with the top axle **608***a* or via other mechanisms, as noted elsewhere herein. In some instances, the motor-generator **310** may be a synchronous reluctance motor that may decouple/freewheel/coast.
(259) As illustrated, multiple internal rotor shunts **972** may be located radially inward from the magnets **952**. In implementations where the magnets **952** are cylindrical, the internal rotor bottom **964** may include contours or other structures for holding the magnets **952** in place.
(260) Although not shown, bar magnets **952** may be coupled with an internal rotor **964** and external rotor **962**. As noted above, the magnetic coupling **318** may include shunts **972** or **974** that hold the magnets **952** and/or direct the magnetic field. In some implementations, an external rotor **962** body and/or internal rotor **964** body may each clamp over and/or around the magnets **952** to hold them in place. For instance, the internal rotor **964** body may include a lip that curves around a face of its magnets to hold them in place.
(261) In the foregoing description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the technology. It will be apparent, however, that the technology described herein can be practiced without these specific details.
(262) Reference in the specification to “one implementation”, “an implementation”, “some implementations”, or “other implementations” means that a particular feature, structure, or characteristic described in connection with the implementation is included in at least one implementation of the disclosure. The appearances of the term “implementation” or “implementations” in various places in the specification are not necessarily all referring to the same implementation.
(263) In addition, it should be understood and appreciated that variations, combinations, and equivalents of the specific implementations, implementations, and examples may exist, are contemplated, and are encompassed hereby. The invention should therefore not be limited by the above-described implementations, implementations, and examples, but by all implementations, implementations, and examples, and other equivalents within the scope and spirit of the invention as claimed.

Claims

1. A system comprising: a flywheel including a rotatable mass component and one or more axles coupled with the rotatable mass component, the one or more axles extending from a top of the rotatable mass component and from a bottom of the rotatable mass component; a bottom bearing assembly coupled with the one or more axles at the bottom of the rotatable mass component; a top bearing assembly coupled with the one or more axles at the top of the rotatable mass component; a support structure coupled with the top bearing assembly and the bottom bearing assembly, the support structure including a base portion and a lid, the lid including a magnetic lift member on a bottom of the lid and a bearing housing on a top of the lid between a motor and the magnetic lift member, the magnetic lift member lifting the rotatable mass component toward the bearing housing; and the motor coupled with the one or more axles at the top bearing assembly.
2. The system of claim 1, wherein: the motor is mounted to the support structure via one or more braces, the one or more braces providing a space between the support structure and the motor, the space being large enough to insert a top bearing between the motor and the support structure without removing the one or more braces.
3. The system of claim 1, wherein the support structure includes an enclosure having a cylindrical shape, the motor being mounted to the enclosure in line with an axis of rotation of the one or more axles.
4. The system of claim 1, wherein: the top bearing assembly coupled with the one or more axles, the top bearing assembly including a top bearing coupling with the one or more axles and a magnet that applies a pulling force to the rotatable mass component.
5. The system of claim 4, wherein the bottom bearing assembly includes a positioning mechanism configured to raise and lower a bottom bearing, the bottom bearing coupling with the one or more axles.
6. The system of claim 1, wherein the support structure includes: an enclosure tub having a bottom and one or more sides coupled with the bottom, the bottom including a perforation, the bottom bearing assembly being located inside the perforation in the bottom, the base portion including the enclosure tub.
7. The system of claim 6, wherein: the lid is coupled with the one or more sides to form a top of the enclosure tub, the lid providing vertical support to the top bearing assembly.
8. The system of claim 7, wherein the top bearing assembly supports more than half of a weight of the rotatable mass component.
9. The system of claim 1, wherein the rotatable mass component includes a plurality of stacked plates.
10. The system of claim 9, wherein the plurality of stacked plates includes two clamping plates and two or more stacking plates located between the two clamping plates, the two clamping plates being pulled together by one or more fasteners to place a compressive force on the two or more stacking plates.
11. The system of claim 10, wherein the one or more axles include a top axle and a bottom axle, the top axle being separated from the bottom axle by the two or more stacking plates.
12. The system of claim 1, wherein: the support structure includes the base portion and the lid, the bottom bearing assembly being coupled with the base portion and the top bearing assembly being coupled with the lid; the motor is coupled with the lid via one or more braces; and the lid includes an accessory mounting plate holding one or more of a vacuum pump and a supercapacitor.
13. The system of claim 12, wherein the lid includes a plurality of ribs separating a body of the lid from the accessory mounting plate, the plurality of ribs merging at a ring at a center of the lid, at least a portion of the top bearing assembly being located within a radius of the ring.
14. The system of claim 12, wherein the magnetic lift member includes: a plurality of magnets mounted to a bottom surface of the lid, the plurality of magnets lifting the rotatable mass component to remove a vertical force from the bottom bearing assembly.
15. The system of claim 1, wherein the support structure includes a plurality of feet around a peripheral edge of the support structure.
16. The system of claim 1, wherein: the support structure includes a cylindrical enclosure, the rotatable mass component being located inside the cylindrical enclosure; the motor and a vacuum pump are mounted on top of the lid; and a plurality of feet are located around a peripheral edge of the cylindrical enclosure.
17. A mechanical-energy storage unit comprising: a flywheel including two clamping plates and two or more stacking plates located between the two clamping plates, the two clamping plates being pulled together by one or more fasteners to place a compressive force on the two or more stacking plates; a top axle coupled with a top clamping plate of the two clamping plates, a bottom axle coupled with a bottom clamping plate of the two clamping plates; a top bearing coupled with the top axle; a bottom bearing coupled with the bottom axle; and a support structure coupled with the top bearing and the bottom bearing, the support structure including a base portion and a lid, the lid including a magnetic lift member on a

bottom of the lid and a bearing housing on a top of the lid between a motor and the magnetic lift member, the magnetic lift member lifting the flywheel toward the bearing housing, the bearing housing holding the top bearing.

18. The mechanical-energy storage unit of claim 17, wherein: the top axle is separated from the bottom axle by the two or more stacking plates; the top axle is held at an axis of rotation by the top bearing; and the bottom axle is held at the axis of rotation by the bottom bearing.

19. The mechanical-energy storage unit of claim 17, wherein the magnetic lift member includes: a plurality of magnets mounted to the support structure, the plurality of magnets lifting the flywheel toward the top bearing.

20. A flywheel assembly comprising: an enclosure having a top, a bottom, and an interior cavity, the top including a lid, the lid including magnetic lift member on a bottom of the lid and a bearing housing on a top of the lid between a motor and the magnetic lift member, the magnetic lift member lifting a flywheel toward the bearing housing; the flywheel located in the interior cavity of the enclosure; a bottom axle coupled with a bottom side of the flywheel; a top axle coupled with a top side of the flywheel, the top axle being a separate component than the bottom axle; a bottom bearing coupling the bottom axle with the bottom of the enclosure; and a top bearing coupling the top axle with the top of the enclosure.